Rewiring the market time clock in Spanish electricity markets: strategic behaviour and market power with higher granularity

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CEMFI Master in Economics and Finance

June 2026

Abstract

Spain's wholesale electricity market moved its intraday auctions to a 15-minute trading grid in March 2025 and the day-ahead market in October 2025, the first European case where both stages of a sequential auction transitioned to a finer trading interval in close succession. We ask whether shorter trading-period granularity reshapes the strategic margin: who uses the within-hour freedom, when in the day, through which equilibrium channel, and whether the reform amplifies or compresses market power.

We build a sequential-market Cournot model with a strategic bidder, a forecast-driven renewable aggregator, and a capacity-bound arbitrageur, augmented with a granularity selector at each stage, and trace the Spanish reform path $(1, 1) \to (1, Q) \to (Q, Q)$. The model delivers eight sign predictions across the reform sequence, including an asymmetric persistence of within-hour wedge dispersion under per-product limited arbitrage.

Identification combines (i) a per-curve difference-in-differences on within-day bid-shape outcomes using flat hours as the within-day control for critical hours, and (ii) a Bayesian counterfactual on level outcomes with year-by-year renewable-coefficient interactions, on rich OMIE bid-level data (every unit, every period, every offer tranche).

Combined-cycle gas day-ahead translates up, flattens, and bends convex at the day-ahead reform (Cournot scale-up); the same fleet's anticipation response at the intraday reform translates the day-ahead curve down, steepens, and bends concave; intraday hydro and pump-storage add tranche density. Demand-side buyers raise their willingness-to-pay at the top of the ladder in step with the sell-side shift, consistent with an equilibrium re-coordination rather than a one-sided supply shift. The within-hour wedge SD jumps at the intraday reform and does not collapse at the day-ahead reform, matching the model's binding per-product arbitrage capacity.

The reforms compress rather than amplify rents in the auction channel: the day-ahead/intraday wedge mean is stable across regimes, and the system-cost decomposition shows the granularity reforms shrink REE's up-direction adjustment channels. The simultaneous growth in Fase I up-redispatch over the same calendar window is driven by the post-blackout reinforced-operation regime, not by the granularity reforms. Granularity is a *strategic-latitude reveal*, not a market-power amplifier.

Contents

Glossary of acronyms

OMIE	Operador del Mercado Ibérico de Energía	PDBC	Programa Diario Base de Casación
REE	Red Eléctrica de España	PDBF	PDBC plus bilateral contracts
CNMC	Comisión Nacional de los Mercados y la Competencia	PIBCI	Programa Intradiario Base de Casación Incremental
ENTSO-E	European Network of TSOs for Electricity	TR	Tiempo Real (real-time technical restrictions)
ESIOS	Sistema de Información del Operador del Sistema	Fase I/II	pre-/post-IDA technical-restriction redispatch
EU	European Union	PO	Procedimientos de Operación
DA	day-ahead auction	aFRR	automatic Frequency Restoration Reserve
IDA	intraday auctions	mFRR	manual Frequency Restoration Reserve
SIDC	Single Intraday Coupling	RR	Replacement Reserve
XBID	cross-border intraday continuous market	BRP	Balance Responsible Party
EPEX	European Power Exchange	BSP	Balance Service Provider
MTU	market time unit (MTU60 hourly; MTU15 15-min)	CCGT	combined-cycle gas turbine
ISP15	imbalance settlement at 15 min (2024-12-11)	RES	renewable energy sources
ID15	intraday at 15 min (2025-03-19)	PV	photovoltaic
DA15	day-ahead at 15 min (2025-10-01)	MIC	minimum-income condition (offer type)
DA60	pre-DA15 hourly day-ahead	HHI	Herfindahl–Hirschman Index
MCP	market clearing price	MW / MWh / GWh	megawatt / megawatt-hour / gigawatt-hour

0 Research abstract paper (RAP) *(to be deleted)*

Positioning. How shorter trading intervals reshape wholesale electricity bidding and prices has been studied for Germany (Märkle-Huß, Feuerriegel and Neumann, 2017) and Australia (Csereklyei and Khezr, 2024). The recent two-step Spanish move to 15-minute trading — intraday auctions in March 2025, day-ahead market in October 2025 — has not yet been evaluated, and it lands in a setting reshaped by the April 2025 blackout and the continuous reinforced-operation regime that followed, requiring a careful empirical strategy to disentangle the reform's effect from the regulatory response to the blackout.

Research Question. Does shorter trading-period granularity reshape the strategic margin in sequential electricity markets? Who uses the within-hour freedom, when during the day is it used, through which equilibrium channel does the reform operate, and does it amplify or compress market power? Do prices fall or rise, and how does the price gap between sequential markets evolve?

Answer. Granularity acts as a *strategic-latitude reveal*: it does not change marginal costs, capacity or who participates; it expands the room for within-hour bid discrimination, and the equilibrium response traces exactly that.

We build a sequential-market Cournot model with a strategic bidder, a forecast-driven renewable aggregator, and a capacity-bound arbitrageur linking the sequential markets, and show that the reform expands within-hour bid-ladder discrimination, pushes prices down on the side of the market that becomes finer (with the other market co-moving down in anticipation), and shifts forecast-driven volumes into the finer market while the average price gap between sequential markets stays at zero. The arbitrageur's bounded capacity prevents the within-hour price-gap dispersion from collapsing, so within-hour heterogeneity persists across both reform steps, and the welfare reading is consumer-favourable on the auction side.

We confirm a subset of these mechanisms in rich Spanish market and bid-level data. Conventional generators widen their within-hour bidding schedules where within-hour residual demand actually varies (the bid-shape DiD of §6.1); the within-hour price-gap dispersion jumps under the first reform and does not collapse at the second, surviving every robustness check we run including the year-varying renewable-coefficient diagnostic (§6.6); the intraday bid stack reveals active position rebalancing by producers across every technology, with the sell-side share rising sharply at the spot reform for nuclear, wind and pump-storage (§6.2, §6.3); and the system-level cost decomposition shows the granularity reforms shrinking the up-direction adjustment-cost channels REE has to call (§6.8). *Level-outcome predictions that do not survive the year-varying renewable-coefficient spec — a symmetric price drop at the spot reform, a day-ahead combined-cycle cleared scale-up at the forward reform, and most per-tech cleared-MW reallocations — we report as consistent in sign with the model but not statistically retained as separate findings.* The channels we do retain compress rather than amplify market power, and the post-blackout cost rise on the same calendar window is a separate regulatory response, not a granularity effect.

1 Introduction

2 Institutional setting: sequential markets, three reforms, and a blackout

2.1 Spanish wholesale architecture: sequential auctions and balancing

Takeaway. Spain runs a sequential auction stack — day-ahead → Fase I restrictions → three intraday auctions → continuous intraday → Fase II → balancing settlement — under uniform-price multi-unit clearing. OMIE auction-cleared programmes are pre-restriction by construction (REE redispatch enters through separate offers), so we can isolate competitive clearing from system-operator interventions. Wholesale concentration is moderate (HHI ∼1 100); ancillary-service concentration is high (HHI >4 000).

The Iberian wholesale market clears in a sequence of auctions operated by OMIE, the market operator. The headline auction is the **day-ahead market** (DA), a uniform-price hourly (since 1 October 2025, quarter-hourly) auction held at noon for the following day's 24 clock-hours. The day-ahead clearing produces the basic day-ahead programme (PDBC), to which traders add their bilateral contract executions to form the functioning day-ahead programme (PDBF). After DA, the system operator REE checks the schedule for network feasibility and runs a **technical-restrictions** process (*Fase I*), which procures up-redispatch or curtailment from specific units through a separate pay-as-bid restriction-offer market (PO 3.2, settlement PO 14.4 ap. 20.1). The post-Fase-I schedule is then re-balanced by participants in the **intraday auctions** (IDA), a sequence of three European auctions cleared at standardised hours through the Single Intraday Coupling (SIDC) mechanism. Between and after the IDAs, the **continuous intraday market** (XBID-coupled) lets participants adjust positions until shortly before delivery. A second restrictions process (*Fase II*) clears after IDA. The balancing market and ex-post settlement close the sequence; the system operator's reference for imbalance accounting is the **imbalance settlement period** (ISP), historically 60 minutes, switched to 15 minutes on 11 December 2024.

Three structural features of this sequence matter for the analysis. First, the day-ahead and intraday auctions both use uniform-price multi-unit clearing, so within-hour discrimination is possible only through the bid stack itself (the price of marginal accepted tranches), not through differential clearing rules. Second, the OMIE auction-cleared programmes (PDBC, PIBCI) are *pre-restriction by construction*: REE's redispatch enters through separate restriction offers, not through the auction clear, so OMIE-cleared volumes track competitive clearing decisions rather than the system operator's restriction call. Third, the relevant strategic margin is per-product within the competing tier of each unit's bid, not the aggregate wholesale clear: market concentration at the wholesale level is moderate (Herfindahl-Hirschman Index around 1 100 over 2016–2024), while the ancillary-services side that REE procures through restrictions is highly concentrated (HHI above 4 000 for up-restrictions; Appendix C.2 collects the evidence).

2.2 The three 15-minute reforms: ISP15, ID15, DA15

Takeaway. Three layered reforms in seven months. ISP15 (Dec 2024): REE's imbalance settlement period moves from 60 to 15 minutes; OMIE contracts stay hourly. ID15 (Mar 2025): intraday auctions and continuous intraday market move from 60 to 15 minutes. DA15 (Oct 2025): day-ahead market follows. Wherever an effect could plausibly attach to more than one reform, we identify the channel separately by window.

Three reforms within seven months changed the market time unit of three different layers of this architecture. **ISP15** (11 December 2024) moved REE's imbalance settlement reference from 60 to 15 minutes. The OMIE auction contract grid was unchanged — DA and IDA still cleared hourly — but the penalty for deviating from a scheduled position now resolved at the quarter-hour rather than the hour, so any within-hour gap between scheduled and realised generation triggered a 15-minute imbalance bill rather than being averaged into the hourly position. **ID15** (19 March 2025) moved the intraday auctions and the continuous intraday market from 60 to 15 minutes: each clock-hour of delivery in IDA now hosted four 15-minute products where it previously hosted one 60-minute product. **DA15** (1 October 2025) extended the move to the day-ahead market on the same logic. A separate prior reform (14 June 2024) replaced the six Iberian (MIBEL) intraday auctions with three European SIDC auctions; the post-IDA-reform three-session architecture is the pre-window for our ID15 analyses.

The reforms are layered rather than independent: ISP15 changes the settlement reference that bidders face in every subsequent auction; ID15 lands on top of the new ISP15 reference; DA15 lands on top of both. Wherever an effect could plausibly be attributed to more than one reform, we report the channel separately — the ISP15 settlement-only wedge SD jump (§6.6) and the ID15 incremental effect on top of it are an example.

2.3 The April 2025 blackout and the reinforced-operation regime

Takeaway. The blackout (28 April 2025) cascaded from voltage- and reactive-power-control failures, not from market microstructure (ENTSO-E Expert Panel). REE's continuous reinforced-operation regime (*operación reforzada*) that followed is a separate regulatory layer: it tightens voltage-control requirements and expands Fase I up-redispatch payments by EUR ∼5M/day, almost entirely on combined-cycle gas. The DA15 BSTS pre-window starts on the blackout date, holding reforzada constant across the cutover by window construction; the ID15 windows sit entirely before the blackout.

On 28 April 2025, the Iberian peninsula suffered a system-wide blackout. The ENTSO-E Expert Panel's final report (ENTSO-E Expert Panel, 2026) attributes the cascade to voltage- and reactive-power-control failures: renewable plants operating at fixed power factor, manually-operated shunt reactors, divergent overvoltage protection settings, converter-driven instability, and absence of power-system stabilisers on several large units. The market microstructure layer is not in the panel's root-cause tree. REE responded by introducing a continuous *reinforced-operation* regime (*operación reforzada*), formally codified in CNMC (2025b) but effectively in force from the blackout date onward. The regime tightens voltage-control requirements on conventional units and substantially expands the use of Fase I redispatch to keep combined-cycle gas plants synchronised for voltage support, even when they would not have cleared in the day-ahead auction. The cost footprint is large: Fase I up-redispatch payments balloon by approximately EUR 5 M per day after the blackout and concentrate almost entirely on combined-cycle gas (Appendix C.4).

Identification consequence. The reforzada regime is continuous from 28 April 2025 onward; it therefore overlaps in calendar terms with the DA15 reform (1 October 2025) but not with the ID15 reform (19 March 2025, with a clean 40-day pre-blackout post-window). The DA15 BSTS analyses use a pre-window that starts on the blackout date (2025-04-28), so reforzada is held constant across the DA15 pre- and post-windows by construction (§5.1). For the ID15 reform there is no reforzada overlap: both the pre- and post-windows sit entirely before the blackout. This window construction is the key identification move that separates the granularity reforms from the reforzada regulatory response — a separation that is essential because the headline ajuste-cost rise across the reform window is reforzada-driven (Fase I up-redispatch growth) rather than MTU15-driven (the granularity channels themselves shrink across the same window; Appendix C.3).

3 Theory: a sequential-market Cournot model with granularity selectors

3.1 Setup: three sequential auctions, three agents

Takeaway. A two-stage Cournot model with a strategic bidder, a forecast-driven renewable aggregator, and a capacity-bound arbitrageur, augmented with a granularity selector $Q^m \in \{1, Q\}$ at each stage. The reform traces the Spanish path $(1,1) \to (1,Q) \to (Q,Q)$ and delivers eight sign predictions for prices, bid ladders, cleared MWh, and within-hour wedge dispersion.

We extend Ito and Reguant (2016)'s sequential-markets framework on two dimensions — an explicit operational renewable aggregator and a granularity variable $Q^m \in \{1, Q\}$ at each stage — and trace the Spanish backward reform path $(1,1) \to (1,Q) \to (Q,Q)$. The mechanism in one line: when a stage becomes more granular, the aggregator shifts MWh into it in proportion to within-hour forecast-error variance (the imbalance penalty is mechanically lower at finer resolution); the strategic bidder sees a per-product rather than pooled residual demand and re-optimises its bid ladder per subperiod; equilibrium prices and cleared MWh re-clear against that finer ladder; and the within-hour wedge dispersion does not collapse to zero because per-product arbitrage capacity binds at K/Q. Full derivations live in Appendix D; the main body collects the comparative-statics output of each state along the path and the eight mechanisms that explain the empirical patterns of §6.

A representative clock-hour contains Q equal-length subperiods. Three sequential stages operate in order: a **forward** auction (Stage 1, $m = F$), a **spot** auction (Stage 2, $m = S$), and ex-post **balancing** settlement (Stage 3, $m = B$). Each stage has its own *granularity* $Q^m \in \{1, Q\}$ — it clears either once per hour or once per subperiod — defining a $2^3 = 8$-state policy space. The main analysis fixes the auction-side path $(Q^F, Q^S) = (1,1) \to (1,Q) \to (Q,Q)$. The forward and spot markets map empirically to the electricity day-ahead and intraday auctions respectively; we use the more general terminology throughout. Figure 1 summarises.

Bidders and primitives. The **strategic bidder** is a residual monopolist with constant marginal cost c and capacity K, which it can deliver *exactly* (no production shock); it therefore does not internalise the imbalance penalty γ. It posts a two-tranche ladder per delivery product to discriminate against the within-hour demand shock μ_t (variance σ^2_μ). The **operational aggregator** is a price-taker with stochastic production $y_t = \bar{y}_t + \eta_t$ (forecast-error variance σ^2_η); it allocates total hourly MWh $\bar{Y}_h$ across forward and spot and *does* internalise the imbalance penalty, which scales with $\gamma(Q^B)$. The **arbitrageur** carries a net position s from forward to spot, in one of the four regimes of Ito and Reguant (§3.2); arbitrage transmits price impact but not imbalance risk. The Ito and Reguant residual-demand specification — forward intercept $\bar{A}_t = \bar{A} + \mu_t$, spot demand inheriting the spread $p_1 - p_2$ with per-subperiod thinness b_{2t}, both net of s and aggregator supply — is stated in full in Appendix D.1; we keep the primitives $(\sigma^2_\mu, \sigma^2_\eta, b_{mt})$ general.

Solution concept. We solve the game by **backward induction**: at Stage 2 the strategic bidder takes the forward position as given and best-responds to spot residual demand; the arbitrageur takes the implied price spread and chooses s within its regime; at Stage 1 the strategic

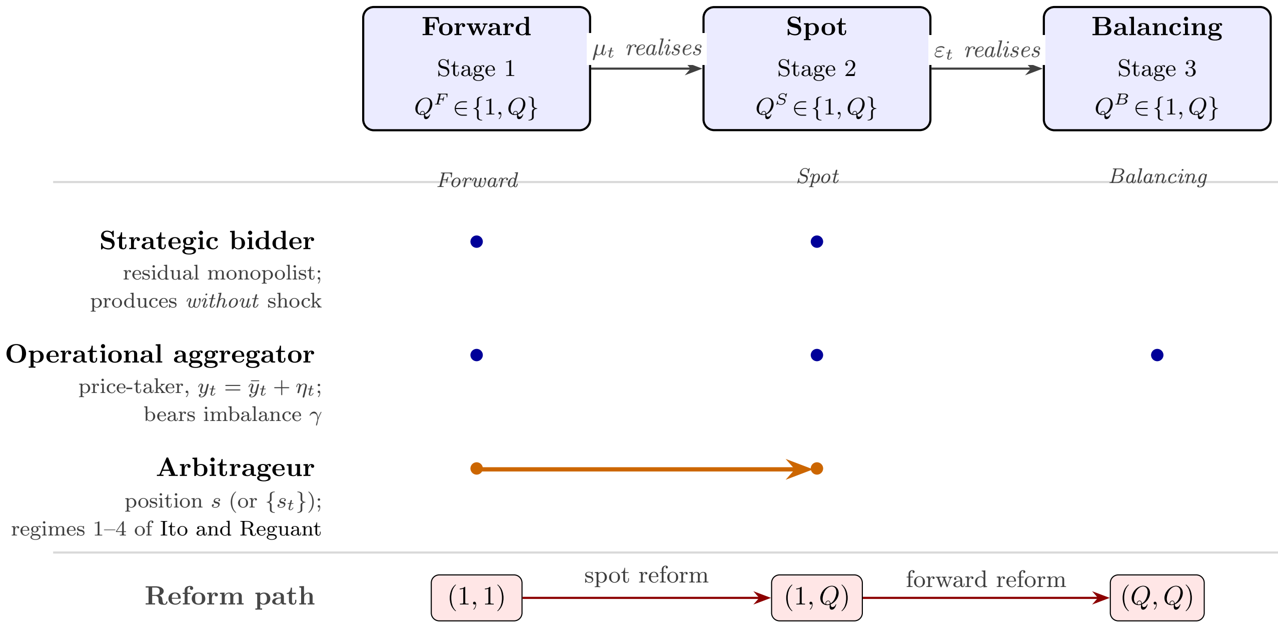


Figure 1: Sequential markets, granularity, and agents. *Top:* a forward market, a spot market, and a balancing settlement on a timeline; each has its own granularity selector $Q^m \in \{1, Q\}$ and information resolves between them. The forward market maps empirically to the electricity day-ahead auction, the spot market to the intraday auction. *Middle:* which agents act at which stages (blue dots), and the arbitrageur position s linking forward to spot (orange arrow). The strategic bidder produces without uncertainty and does not internalise the imbalance penalty; only the operational aggregator does, since it bears η_t. *Bottom:* the auction-side reform path $(Q^F, Q^S) = (1, 1) \to (1, Q) \to (Q, Q)$ traversed in the main analysis.

bidder commits its forward ladder using rational expectations of the Stage-2 best responses. Full step-by-step backward induction in Appendix D.4–D.6.

Model abstractions: residual monopolist, Cournot vs supply-function, and the role of the balancing stage. Three simplifications deserve flagging. First, the strategic bidder is a single residual monopolist with constant marginal cost c and capacity K, even though Spanish wholesale concentration is moderate (HHI ∼1 100) and several firms compete on the bid stack. We use the residual-monopolist as a stylised device that delivers qualitative comparative statics on within-band conduct. Second, we use Cournot best responses rather than supply-function equilibria; the latter is the textbook equilibrium concept for uniform-price multi-unit auctions (Klemperer and Meyer, 1989) but in a sequential-market setting it gets intractable. Third, the balancing stage in the model is a passive settlement of the residual imbalance, *not* an active auction with bidding. The aggregator internalises its own deviation through the penalty γ; the strategic bidder produces with certainty and so carries no individual imbalance; the system-side residual on the demand leg (μ_t unabsorbed by hourly markets) collects in C^I_{sys} for the welfare account (Appendix D.7, D.9). In reality REE runs aFRR/mFRR auctions where firms bid actively; we abstract from that to keep the focus on how forward/spot granularity reshapes the auction stages. The model's prediction that finer auctions absorb shocks upstream and shrink the balancing residual is what §6.8 reads against the empirical ajuste-cost decomposition.

Renewable fringe and our relation to Ito and Reguant. Ito and Reguant's residual-demand interpretation already absorbs a renewable (or thermal) competitive fringe into the slope b_m: the strategic bidder serves $A - b_m p_m$, the implicit fringe serves the complement. Their setup is sufficient to deliver the strategic-bidder, arbitrage, and price-wedge mechanisms but is silent on *why* the fringe's willingness to supply changes with granularity. We make the fringe explicit (the operational aggregator with σ_η^2 and an imbalance penalty γ) so the quantity-shift channel (M1) can be derived from first principles rather than absorbed into a granularity-dependent slope $b_m(Q^S, \sigma_\eta^2)$.

3.2 Arbitrage regimes: four benchmarks, one empirical

Following Ito and Reguant (2016, §I.A and Table 1), arbitrage profit per matched product is $\pi^A(s) = s\,(p_1 - p_2)$. Four regimes pin s: TO EXTEND THIS!!!

(1) *No arbitrage*: $s = 0$, positive forward premium $p_1 > p_2 > c$.

(2) *Full / competitive arbitrage*: s closes the spread, $p_1 = p_2$; the strategic bidder withholds more in response.

(3) *Limited arbitrage*: a capacity ceiling $s \leq K$ binds, sustaining a positive wedge whose size depends on K.

(4) *Strategic arbitrage*: the arbitrageur internalises price impact and stops short of full, giving the closed-form benchmark

$$\boxed{p_1 - p_2 \;=\; \frac{p_1 - c}{4} \;>\; 0.} \tag{1}$$

Comparative-statics signs are the same across (1), (3), (4) (Ito and Reguant Table 1). AVOID TALKING ABOUT EMPIRICS HERE!

3.3 Pre-reform $(1,1)$: scarcity wedge, no within-hour dispersion

With both auctions hourly, no within-hour wedge dispersion is mechanically feasible — each market has one product per clock-hour. Two features carry through every subsequent reform: the Ito and Reguant scarcity wedge $p_1 > p_2 > c$ survives under any non-full-arbitrage regime, and both within-hour shocks (μ_t on demand, η_t on aggregator production) load onto the balancing residual because neither hourly market can absorb them.

Both auctions hourly. The Ito and Reguant scarcity wedge $p_1 > p_2 > c$ obtains; neither the hourly forward nor the hourly spot can absorb within-hour shocks, so the balancing discrepancy is $\Delta_t = \mu_t + \eta_t + \varepsilon_t$.

3.4 Asymmetric state $(1, Q)$: four spot-side mechanisms

Takeaway. When only the spot becomes finer (Spain's ID15 state), four mechanisms activate on the spot side: M1 aggregator quantity shifts to the granular market; M2 granular-side price drops; M3 within-hour wedge dispersion emerges from per-product thinness asymmetry; M4 spot-side bid ladders widen. The forward market is unchanged so μ_t still loads onto balancing.

Spot granularity simultaneously activates four mechanisms: (M1) the aggregator shifts quantity to the granular market, in proportion to forecast-error variance; (M2) the granular-side price drops in response; (M3) within-hour wedge dispersion emerges from per-subperiod thinness asymmetry; and (M4) the strategic bidder's spot-side two-tranche ladder widens. The forward market is unchanged, so the balancing discrepancy still carries μ_t.

The spot market becomes per-period; the forward market stays hourly.

(M1) The aggregator shifts quantity to the granular market. Per-period spot scheduling matches per-subperiod expected production, collapsing within-hour imbalance to residual noise;

the hourly forward cannot adjust per subperiod and incurs a mismatch cost. The aggregator therefore shifts Δ^* MWh from forward to spot, and **the shift is larger when within-hour forecast uncertainty is larger**:

$$\boxed{\partial\Delta^*/\partial\sigma_\eta^2 \;>\; 0.} \tag{2}$$

Derivation in Appendix D.2.

(M2) The granular-side price drops. More aggregator supply on spot reduces the strategic bidder's spot residual demand. The equilibrium spot price drops on average by an amount of order Δ^*/b_2; through the aggregator channel alone, the forward price drifts up modestly as the strategic bidder serves a larger forward share. Two channels operate simultaneously on the forward side, however, and they cut in opposite directions. The Stage-1 option-value channel introduced for the symmetric state in §3.5(v) is already active here: by backward induction, once Stage-2 becomes finer the Stage-1 bidder anticipates the spot-side ladder widening of (M4) and restructures its forward bid accordingly, which can push the forward price down. The empirical DA-side drop under ID15 (§6.5) is consistent with the anticipation channel dominating the aggregator channel on the forward leg. The net cross-market effect remains positive-sum.

(M3) Block parity reveals within-hour spot dispersion. The hourly forward can be arbitraged only against the *average* of spot per-period prices — under full arbitrage, $p_1 = (1/Q)\sum_t p_{2t}$ (block parity); under strategic arbitrage, the same average condition plus the Ito and Reguant wedge $(p_1 - c)/4$. This leaves room for within-hour spot dispersion whenever per-subperiod thinness b_{2t} varies across products: thinner products carry higher clearing prices for the same aggregate block. The wedge SD inherits this thinness asymmetry directly, attenuated by half under strategic vs full arbitrage (Appendix D.5).

(M4) Spot-side bid ladders widen. Per-period clearing rewards two-tranche discrimination across products: the strategic bidder posts a low tranche for low-μ_t states and a high tranche for high-μ_t states. The optimal pre-realisation spread is

$$\boxed{\Delta p_{2t}^* \;=\; \frac{\Delta_\mu}{2\,b_{2t}},} \tag{3}$$

strictly larger than the hourly counterpart (Appendix D.3). Empirical measures of within-product bid-shape dispersion (e.g. MW-weighted price SDs, effective tranche counts) are motivated by this widening but are not derived from the two-tranche reduction directly; we use them as empirical proxies, not as model objects.

Balancing discrepancy unchanged at $\Delta_t = \mu_t + \varepsilon_t$: the forward is still hourly so μ_t still loads onto balancing.

3.5 Symmetric state (Q, Q): limited per-product arbitrage

Takeaway. When the forward also becomes per-period (Spain's DA15 state), six mechanisms operate. The empirically critical one is per-product limited arbitrage: per-product capacity binds at $K_t \approx K/Q$, so the within-hour wedge SD does NOT collapse to the strategic-arbitrage benchmark but stabilises at $\sigma_\mu/(4b_1)$ — exactly twice the unlimited strategic benchmark. This is the model-internal account of the empirical asymmetric-persistence finding (§6.6). The Cournot scale-up of per-clock-hour cleared MWh and a cross-market option-value substitution of bid shapes complete the symmetric-state predictions.

When the forward also becomes per-period, six mechanisms operate; the empirically critical one is per-product limited arbitrage. Per-product arbitrage capacity binds at $K_t \approx K/Q$, so the within-hour wedge SD does not collapse to the strategic-arbitrage benchmark but stabilises at $\sigma_\mu/(4b_1)$ plus thinness asymmetry — exactly twice the unlimited strategic benchmark and consistent with the empirical asymmetric-persistence finding. The Cournot scale-up of per-clock-hour cleared MWh and the cross-market option-value substitution of bid shapes complete the symmetric-state predictions.

The forward market also becomes per-period.

(i) Aggregator allocation equalises. The forward-vs-spot hedging differential closes once both markets accept per-subperiod schedules.

(ii) Per-product arbitrage — and arbitrageur market power. The full-arbitrage benchmark gives $p_{1t} = p_{2t}$ per product (wedge SD zero); the strategic-arbitrage benchmark gives the Ito and Reguant wedge (1) per product, collapsing the wedge SD to $\sigma_\mu/(8\,b_1)$. *Both benchmarks rely on the arbitrageur being able to build any per-product position, and both fail empirically.* Arbitrage capacity K is bounded; with Q matched products per clock-hour, per-product capacity is $K_t \approx K/Q$, which binds tightly. The symmetric state is therefore in the **limited-arbitrage regime** (Ito and Reguant Result 3) applied per product: the per-product wedge becomes $(p_{1t} - c)/2$ rather than $(p_{1t} - c)/4$, doubling the wedge SD to $\sigma_\mu/(4\,b_1)$ (Appendix D.6.4); if per-period thinness b_{2t} also varies across products this contributes an additional thinness-asymmetry component. The wedge SD therefore does *not* collapse from the asymmetric-state magnitude. *This is the model-internal structural account of empirical asymmetric persistence: arbitrageur market power generates the non-collapse. Residual dispersion the model does not pin down may reflect additional channels (e.g. behavioural inertia or renewable-penetration trends not absorbed by the covariates); the structural and residual accounts are complementary, not exclusive.*

(iii) Forward-side bid ladders widen. The same mechanism as (M4) on the forward side: optimal spread $\Delta p^*_{1t} = \Delta_\mu/(2b_1)$.

(iv) Cournot scale-up from per-period thinness. Per-period clearing thins the per-product residual demand: each per-subperiod product has fewer competing units than the hourly aggregate, so $b_{1t} < b_1$ on average. Cournot quantity is decreasing in elasticity, so per-product cleared MWh exceeds $1/Q$ of the hourly value and total hourly cleared MWh rises. The condition is $\sum_t b_{1t} < Q\,b_1$ (Appendix D.6.5); the scale-up concentrates in subperiods where per-period thinness drops most relative to the hourly aggregate. No external withholding mechanism is needed — this is the standard Cournot quantity comparative static.

(v) Cross-market bid-shape substitution via Stage-1 option value. Stage-1 forward bidding is optimised by backward induction against expected Stage-2 outcomes. Per-period spot makes Stage-2 profit convex in μ_t (per-period Cournot gives $\partial^2\pi_{2t}/\partial\mu_t^2 = 1/(2b_{2t})$), so Stage-2 expected profit acquires a positive variance correction $\sigma_\mu^2/(4b_{2t})$. The Stage-1 bidder responds by widening the forward ladder to capture both sides of the variation it anticipates downstream. Symmetrically, once the forward is per-period the within-hour discrimination is absorbed at Stage 1 directly and the residual spot ladder tightens. (Derivation in Appendix D.6.6.)

(vi) Balancing efficiency. The per-period forward schedule absorbs μ_t, so only the residual post-Stage-2 noise loads onto the balancing reference: $\Delta_t = \varepsilon_t$.

3.6 Comparative statics across the reform path

Takeaway. Seven sign predictions (P1–P7) across the reform path, of which the asymmetry of activation across the two reform steps is the most distinctive: aggregator channels (M1, M2, spot-side ladder widening) activate at the spot reform; strategic-bidder channels (Cournot scale-up, forward-side ladder widening) activate at the forward reform. The two-agent typology gives a one-to-one match with the reform-by-reform incidence we observe empirically.

The model delivers seven comparative-statics predictions across the three states $(1,1) \to (1,Q) \to (Q,Q)$, summarised in Table 1. The wedge-SD row reports the empirically relevant limited-arbitrage benchmark; the full and strategic benchmarks are derived in eq. (18) for comparison.

Table 1: Comparative statics across the reform path under limited per-product arbitrage (empirical regime). Imbalance row reports $E[\Delta_t^2]$ for the quadratic-cost penalty (§3.7); under uniform $\mu_t \sim U[-\Delta_\mu, \Delta_\mu]$ and $\eta_t \sim U[-\Delta_\eta, \Delta_\eta]$, $E[\mu_t^2] = \Delta_\mu^2/3$ and $E[\eta_t^2] = \Delta_\eta^2/3$.

Outcome	$(1,1)$	$(1,Q)$	(Q,Q)
Aggregator fwd→spot shift	0	$\Delta^*(\sigma_\eta^2) > 0$	equalised
Spot ladder spread Δp_{2t}^*	—	$\dfrac{\Delta_\mu}{2b_{2t}}$	$\dfrac{\Delta_\mu}{2b_{2t}}$
Forward ladder spread Δp_{1t}^*	—	via Stage-1 option	$\dfrac{\Delta_\mu}{2b_1}$
Wedge SD (limited arb.)	0	$\dfrac{K\lvert b_{22}-b_{21}\rvert}{8b_{21}b_{22}}$ $(Q=2)$	$\dfrac{\sigma_\mu}{4b_1}$ + thin. asym.
Cleared MWh / clock-hour	baseline	forward unchanged	$+\dfrac{c(Qb_1 - \sum_t b_{1t})}{2}$
Balancing Δ_t	$\mu_t + \eta_t + \varepsilon_t$	$\mu_t + \varepsilon_t$	ε_t
Imbalance cost $C^I/(\gamma Q)$, quadratic	$(\Delta_\mu^2 + \Delta_\eta^2)/3$	$\Delta_\mu^2/3$	0

Arbitrage regime is empirical. Real arbitrageurs have market power and bounded balance sheets: the persistent forward premium observed across electricity markets rules out full arbitrage, and the per-product capacity constraint binds at (Q,Q). The relevant Ito and Reguant column is Result 3 (limited), not Result 4 (strategic, unlimited); per-product limited arbitrage at (Q,Q) delivers the model's account of the empirical asymmetric-persistence of wedge dispersion (§3.5(ii)).

Predictions to test empirically. The model delivers seven sign predictions across the reform path; §6 verifies them in turn: (P1) quantity shifts from forward to spot at the spot reform, larger in clock-hours with bigger forecast uncertainty; (P2) the granular-side equilibrium price drops in response; (P3) within-hour wedge dispersion emerges from per-subperiod thinness asymmetry (largest in ramp hours); (P4) the strategic bidder's two-tranche ladders widen on the granular side; (P5) total cleared MWh per clock-hour rises at the symmetric state; (P6) within-hour wedge SD does not collapse at (Q,Q) — the empirical asymmetric-persistence finding; (P7) production-unit arbitrageurs exist and are capacity-constrained (we test this directly by checking whether production plants submit both buy and sell offers in OMIE data).

Asymmetry of activation across the two reform steps. The mechanisms above do not all activate at the same reform step, and that asymmetry sits at the centre of the two-agent structure. The aggregator-side channels (M1, M2, the spot-side ladder widening of M4) activate at the *spot* reform (1,1)→(1,Q), because the granularity gain to the aggregator is largest when the spot becomes finer (proximity to delivery is where the within-hour forecast variance σ_η^2 matters most). The strategic-bidder Cournot scale-up of (iv) at (Q,Q) and the forward-side ladder widening of

(iii) activate only at the *forward* reform (1,Q)→(Q,Q), because they require a per-product residual demand on the forward leg. The model therefore predicts that the operational agent's footprint is largest at the spot reform and the strategic agent's footprint is largest at the forward reform, even though both reforms move equilibrium prices and quantities.

3.7 Welfare across reforms and arbitrage regimes

Takeaway. Three positive welfare channels (imbalance saving, Cournot scale-up, less strategic withholding under limited per-product arbitrage) and one distributional channel (bid-ladder rent, mostly CS→producer transfer). All three positive channels operate at every reform step; total welfare is unambiguously up under both reforms.

Under uniform shocks $\mu_t \sim U[-\Delta_\mu, \Delta_\mu]$ and $\eta_t \sim U[-\Delta_\eta, \Delta_\eta]$, the reform's welfare effect decomposes into three positive channels and one distributional one:

- **Imbalance saving (dominant gain).** The spot reform absorbs the aggregator's η_t at delivery; the forward reform absorbs the demand-side μ_t at scheduling. Total imbalance cost falls from $\gamma Q(\Delta_\mu^2 + \Delta_\eta^2)/3$ at $(1,1)$ to $\gamma Q\,\Delta_\mu^2/3$ at $(1,Q)$ to 0 at (Q,Q).
- **Cournot scale-up at (Q,Q).** Per-period thinness ($\sum_t b_{1t} < Qb_1$) delivers $(c/2)(Qb_1 - \sum_t b_{1t})$ additional MWh at the pre-reform mark-up, a positive welfare gain.
- **Less strategic withholding under limited per-product arbitrage.** The spot markup falls from $3(p_1 - c)/4$ (strategic) to $(p_1 - c)/2$ (limited per product), per Ito-Reguant Result 2. The empirical asymmetric-persistence of wedge SD characterises the limited regime; it is not a welfare cost.
- **Bid-ladder rent (distributional).** The richer two-tranche ladder is mostly a CS-to-π^{SB} transfer.

All three channels are positive at every reform step. Per-component derivations under quadratic imbalance penalty $\gamma\mathcal{I}^2$ in Appendix D.9.

4 Data: OMIE bid-level offers, weather, redispatch

4.1 Sources

Takeaway. Bid-level OMIE data (every unit, every period, every offer tranche) covering 2024-06-14 to 2026-03-06, plus ENTSO-E load and renewable generation, plus ESIOS daily settlement of REE's adjustment-cost channels. The bid-level granularity is what lets Spec C identify within-day shape changes; the system-level data feeds Spec A's counterfactual and the cost decomposition.

We combine three sources at daily resolution over 2022-01-01 to 2026-04-30 (with bid-level detail in 2024-06-14 to 2026-03-06, the post-IDA-reform window). **OMIE** provides every bid-level offer in the day-ahead (PDBC) and intraday auctions (PIBCI) markets — one row per (unit, date, period, tranche), with bid price, incremental MW, and structural type (simple / minimum-income condition / block) per OMIE spec v1.37. **ENTSO-E** Transparency Platform provides 15-min actual total load (A65) and per-technology actual generation (A75, codes B16 wind onshore, B18 wind offshore, B19 solar), used to construct the residual-demand series that calibrates the critical/flat hour-class partition (§4.5) and the wind/solar covariates for Spec A. **ESIOS** provides daily settlement of REE's adjustment-cost (ajuste) components — the up-direction channels (TR-up, aFRR-up, mFRR-up, Fase I up-redispatch) and the down-direction channels (Fase II, RR-dn) used in the system-cost decomposition (§6.8).

4.2 Bid offers, reform path, and notation

Indices. Throughout: u unit, d date, p market period (a clock-hour pre-MTU15, a 15-minute quarter post-MTU15), $h \in \{1, \dots, 24\}$ clock-hour, $q \in \{1, \dots, 4\}$ within-hour quarter (post-MTU15 only), s IDA session, t technology, $m \in \{\mathrm{DA}, \mathrm{IDA}\}$ market, k tranche of a step ladder, T_r reform cutover date, $\mathcal{X} \in \{\text{morning ramp}, \text{midday}, \text{evening ramp}, \text{flat}\}$ hour-class.

Bid offer. Each (u, d, p) sell offer is a step ladder $\{(p_k, \Delta q_k)\}_{k=1}^{K_{u,d,p}}$ sorted by ascending price, with p_k the bid price (EUR/MWh) and Δq_k the incremental MW added to the unit's step supply at p_k.[1] Offers carry one of three structural types: *simple* (a bare price–quantity ladder), *minimum-income condition* (MIC), or *block* (all-or-nothing across periods). Most technologies use simple bids; CCGT is the outlier with the bulk of its offers in MIC or block form.[2]

Market clearing. $\mathrm{MCP}_{d,p}$ is the OMIE clearing price (EUR/MWh) of the auction in which the offer was submitted; $Q^{\star}_{d,p}$ is the system cleared quantity. Per-tech cleared MWh per clock-hour: $Q^{\mathrm{cl}}_{t,m,d,h}$.

Reform path. The Spanish wholesale market underwent three granularity reforms in seven months: **ISP15** (2024-12-11), REE's settlement period moved from 60 to 15 min while the OMIE contract grid stayed at 60 min; **ID15** (2025-03-19), intraday auctions and continuous market moved from 60 to 15 min; **DA15** (2025-10-01), day-ahead moved from 60 to 15 min. The April 2025 blackout (2025-04-28) triggered a continuous *operación reforzada* regime (CCGT-led voltage support, post-DA redispatch) that overlaps DA15 calendar-wise but operates on a different mechanism; DA15 windows are constructed so this regime is held constant on both sides of the cutover. Figure 2 maps the windows that partition the sample.

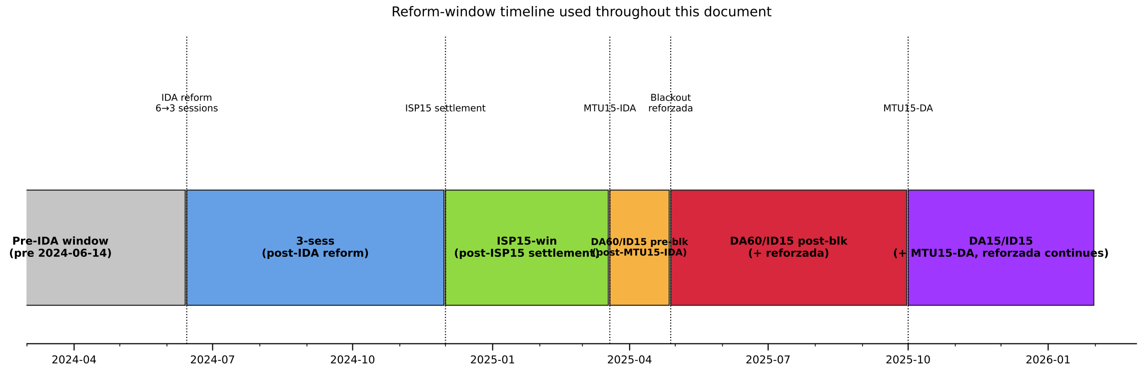


Figure 2: Reform-window timeline. Six named windows partition the 2022-2026 sample at the five reform dates. *Pre-IDA*: baseline 6-session IDA, hourly. *3-sess*: post-IDA-reform 3-session IDA, hourly. *ISP15-win*: REE settlement 15-min, OMIE grid still 60-min. *DA60/ID15 pre-blk*: 40-day pre-blackout window with quarter-hourly intraday. *DA60/ID15 post-blk*: post-blackout reforzada, quarter-hourly intraday. *DA15/ID15*: full 15-min grid post-2025-10-01.

4.3 In-band region and bid-shape outcomes

Takeaway. The bandwidth $\delta = p_{90} - p_{50}$ of the contemporary MCP distribution isolates the *competing tier* of each ladder — tranches close enough to clearing to be in real contention. Per in-

[1] OMIE *Potencia* field, *ficheros OMIE v1.37* §5.1.4.2. Cumulative supply at any price p is $S_{u,d,p}(p) = \sum_{k:\, p_k \le p} \Delta q_k$.

[2] Approximate sell-offer shares by structural type: wind, solar, hydro and pump-storage $\sim 100\%$ simple; nuclear 64% simple, 36% MIC; CCGT 18% simple, 50% MIC, 32% block. See Appendix A.3.

band curve we compute four jointly informative geometric statistics: intercept $\hat{\alpha}$ (level), linear slope $\hat{\beta}$, curvature $\hat{\gamma}$ (convex > 0 / concave < 0), and effective tranche count N_{eff} (inverse-Herfindahl of MW weights).

Bandwidth. Let δ be a bandwidth that isolates the tranches in contention — close enough to the clearing price to compete in the *competing tier* of the offer (CCGT's two-tier structure makes this distinction explicit, Appendix C). We anchor δ to the contemporary contested margin: $\delta = p_{90} - p_{50}$ of the MCP distribution in the analysis window, market by market. The eight bandwidths (one per reform $\times$ {real, placebo} $\times$ {DA, IDA}) are reported below; wider-δ robustness in Appendix A.5.

Window	δ in DA (EUR/MWh)	δ in IDA (EUR/MWh)
ID15 real	50	62
ID15 placebo (2024)	45	46
DA15 real	50	58
DA15 placebo (2024)	45	49

Table 2: Bandwidths δ used to define the in-band region. δ is set to $p_{90} - p_{50}$ of the MCP distribution in each (window, market). Window-and-market-specific by construction. Bandwidth robustness in Appendix A.5.

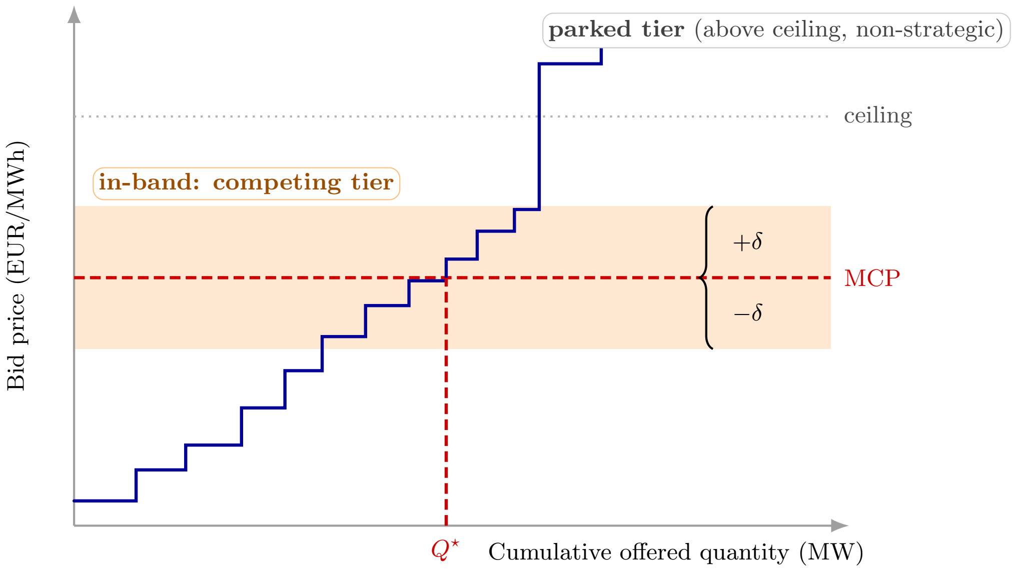


Figure 3: Bandwidth δ and the in-band region. The blue step function is the sorted sell-bid ladder for one quarter. Only tranches falling within $\pm\delta$ of the clearing price (orange band) enter the bid-shape outcomes; the parked tier above the clearing-price ceiling is excluded (Appendix C).

In-band set. For each (u, d, p) curve:

$$\mathcal{B}_{u,d,p} = \{k : |p_k - \text{MCP}_{d,p}| \le \delta\}, \qquad w_k = \frac{\Delta q_k}{\sum_{j \in \mathcal{B}_{u,d,p}} \Delta q_j}, \qquad \bar{p}_{u,d,p} = \sum_{k \in \mathcal{B}} w_k \, p_k.$$

Representative bid curves per technology in critical and flat hours are in Appendix C.4 (Figures 11–12); they illustrate the dispatchable-ladder vs renewable-block contrast.

Four bid-shape outcomes. Per curve, the in-band tranches deliver four non-redundant summaries.

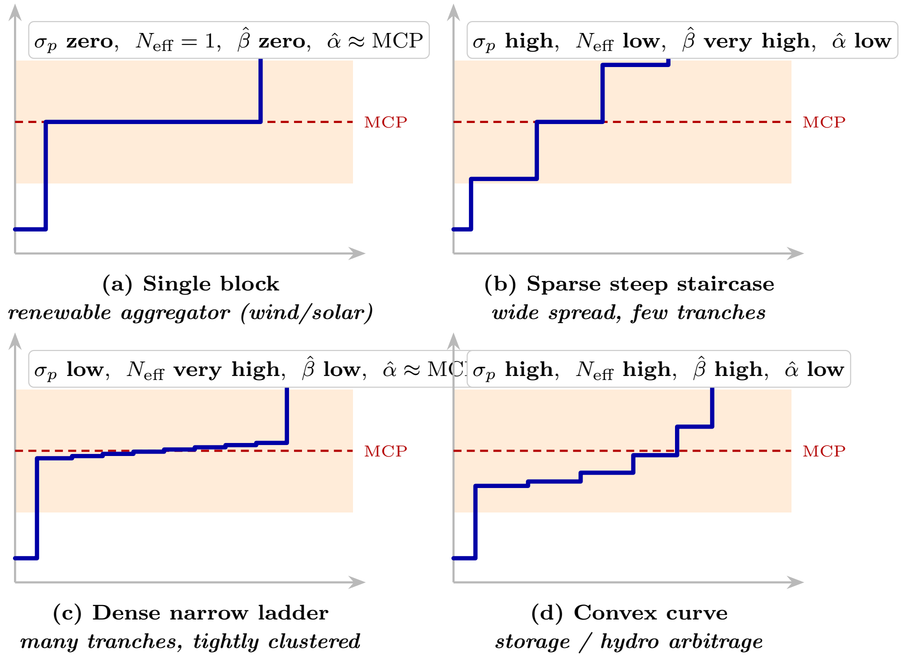


Figure 4: Four stylized in-band bid shapes spanning the $(\sigma_p, N_{\text{eff}}, \hat\beta, \hat\alpha)$ space. Each panel shows one bid ladder against a common MCP (red dotted) and bandwidth δ (orange band), with the implied outcome values noted in the corner.

Level, slope, curvature (unweighted OLS on in-band step endpoints). Let $Q_k = \sum_{j \le k} \Delta q_j$ be the cumulative in-band MW after the kth tranche. Fit a quadratic

$$p_k \;=\; \hat\alpha_{u,d,p} + \hat\beta_{u,d,p}\, Q_k + \hat\gamma_{u,d,p}\, Q_k^2 + e_k$$

by OLS with each in-band tranche contributing one data point with equal weight (no MW weighting). $\hat\alpha$ is the geometric level of the ladder (predicted price at $Q = 0$). $\hat\beta$ is the linear slope (EUR/MWh per MW) and is the direct empirical counterpart of the model's per-product residual-demand slope b_{mt} (§3.1). $\hat\gamma$ is the curvature: $\hat\gamma > 0$ means the curve bends *convex* (slope steepens with Q, the bidder packs cheap MW at the bottom and asks higher prices for the marginal MW); $\hat\gamma < 0$ means *concave* (steep climb early then a flatter tail). With $K = 2$ in-band tranches only $\hat\alpha, \hat\beta$ are identified; $\hat\gamma$ requires $K \ge 3$.

Discrete tranche density.

$$N_{\text{eff},\,u,d,p} \;=\; \Big(\sum_{k \in \mathcal{B}} w_k^2\Big)^{-1}, \qquad w_k = \frac{\Delta q_k}{\sum_{j \in \mathcal{B}} \Delta q_j}.$$

Inverse-Herfindahl of MW weights (1 = a single block, large = many equal-size tranches). N_{eff} captures the granularity of the discrete decisions the bidder actually placed; $(\hat\alpha, \hat\beta, \hat\gamma)$ capture the smooth shape that fits those decisions.

What each outcome captures. The four outcomes are jointly informative and non-redundant: $\hat\alpha$ is level, $\hat\beta$ is linear slope, $\hat\gamma$ is curvature (convex/concave), N_{eff} is discrete density. Reading them together disentangles where the curve sits, how steep it is, whether it bends, and how many discrete decisions it contains. None depends on price-level seasonality since all are computed relative to the contemporaneous MCP via the bandwidth δ. Figure 4 illustrates the outcomes on stylised ladders. We also report the MW-weighted in-band price SD $\sigma_p = [\sum_k w_k (p_k - \bar p)^2]^{1/2}$ as an appendix diagnostic; it is mechanically dominated by $(\hat\alpha, \hat\beta, \hat\gamma)$ once the quadratic fit is in hand and adds no independent geometric content.

4.4 Aggregate outcomes

Takeaway. Five system-level outcomes built off the bid-stack and the cleared prices: in-band offered energy per (tech, market, day, hour-class); the day-ahead/intraday wedge $\Delta_{d,h}$ and its within-day SD Δ_d^{SD}; the intraday-auction in-band sell-share by tech; system ajuste cost (per day, EUR M); and five post-MTU15-only within-hour dispersion outcomes $D^{\bullet}_{d,h}$ that summarise across-quarter shifts in price level, volume, system MCP, ladder spread, and tranche count. The wedge SD Δ_d^{SD} and the IDA in-band sell-share are the load-bearing system-level outcomes for Spec A.

In-band offered energy (per (tech, market, day, hour-class)). Per-curve in-band MW: $m_{u,d,p} = \sum_{k\in\mathcal{B}_{u,d,p}} \Delta q_k$. Aggregated to tech t, market m, hour-class $\mathcal{X}$, and converted to energy with τ_p the period length in hours ($\tau_p = 1$ pre-MTU15, $\tau_p = 1/4$ post-MTU15):

$$E^{\mathrm{band}}_{t,m,d,\mathcal{X}} \;=\; \frac{1}{|\mathcal{X}|}\sum_{h\in\mathcal{X}}\sum_{p\in h}\sum_{u\in t} m_{u,d,p}\cdot\tau_p \qquad \text{(MWh per class-hour)}.$$

The τ_p factor makes the post-MTU15 quarter-sum invariant to the period count; the $1/|\mathcal{X}|$ normalisation makes levels comparable across hour-classes of different sizes.

Cross-market wedge (per clock-hour). Let $\mathrm{MCP}^{\mathrm{DA}}_{d,h}$ and $\mathrm{MCP}^{\mathrm{IDA}}_{d,h}$ be the within-market means of the OMIE clearing prices over whatever sub-hour structure each market has on date d (DA collapses four quarters post-DA15; IDA collapses three sessions and, post-ID15, four quarters):

$$\Delta_{d,h} \;=\; \mathrm{MCP}^{\mathrm{DA}}_{d,h} - \mathrm{MCP}^{\mathrm{IDA}}_{d,h}, \qquad \Delta_d^{\mathrm{SD}} \;=\; \mathrm{sd}_h(\Delta_{d,h}).$$

$\Delta_{d,h}$ is the DA−IDA gap (EUR/MWh) at clock-hour h; Δ_d^{SD} summarises how that gap varies across the 24 clock-hours of day d. The clock-hour collapse makes $\Delta_{d,h}$ directly comparable across regimes (it does not widen mechanically when IDA gains a quarter dimension).

IDA in-band sell-share (per (tech, day)). Restricted to in-band tranches:

$$\mathrm{SS}_{t,d} \;=\; \frac{\sum_{q,k\in\mathcal{B}} q^{\mathrm{sell}}_{t,d,q,k}}{\sum_{q,k\in\mathcal{B}}\bigl(q^{\mathrm{sell}}_{t,d,q,k} + q^{\mathrm{buy}}_{t,d,q,k}\bigr)}.$$

Captures the position of production units between sell-heavy (rent-capture) and buy-heavy (rebalancing) participation in the intraday auction.

System ajuste cost (per day, EUR/M). REE's adjustment-cost components from ESIOS: up-direction (TR-up, aFRR-up, mFRR-up, Fase I) and down-direction (Fase II, RR-dn), stacked positive in Figure 7.

Within-hour dispersion outcomes (post-MTU15 only). For each (unit, date, clock-hour) with four quarters $q = 1, \ldots, 4$:

$$D^{\bar{p}}_{d,h} = \mathrm{sd}_q(\bar{p}_q), \quad D^{m}_{d,h} = \mathrm{sd}_q(m_q)/\bar{m}, \quad D^{\mathrm{MCP}}_{d,h} = \mathrm{sd}_q(\mathrm{MCP}_q), \quad D^{\sigma_p}_{d,h} = \mathrm{sd}_q(\sigma_{p,q}), \quad D^{N_{\mathrm{eff}}}_{d,h} = \mathrm{sd}_q(N_{\mathrm{eff},q}).$$

The five $D^{\bullet}_{d,h}$ outcomes capture across-quarter shifts in price level, volume, system MCP, ladder spread and tranche count. All are mechanically zero under MTU60 (one quarter per clock-hour), so any positive value post-MTU15 is a direct trace of within-hour discrimination. These are used in §4.5 to characterise where within-hour discrimination concentrates.

4.5 Critical / flat hour-class partition

Takeaway. Granular pricing has economic content only where within-hour residual demand actually varies. Measured by 15-minute residual-demand SD, clock-hours partition into critical (morning ramp 5–8h, evening ramp 16–22h; SD 522–1303 MW), flat (overnight 1–3h; SD 336–357 MW), and midday (11–14h; SD 468–502 MW, kept for falsification only). Spec C uses critical-vs-flat as the within-day identification contrast.

Intuition. Granular pricing only has economic content where within-hour residual demand[3] actually varies; if a clock-hour is internally flat, the four quarters look the same and there is nothing to discriminate on. We measure within-hour variation by $\sigma_{\text{within}}(h,d) = \text{sd}_q[\text{load}-\text{wind}-\text{solar}]_{q,h,d}$ on 15-min ENTSO-E data (A65 load; A75 PSR codes B16, B18, B19 for wind+solar) and partition clock-hours by median σ_{within} post-2024-06-14:

- **Critical** $\mathcal{C} = \mathcal{C}_M \cup \mathcal{C}_E$ with $\mathcal{C}_M = \{5,\dots,8\}$ (morning ramp) and $\mathcal{C}_E = \{16,\dots,22\}$ (evening ramp); $\sigma_{\text{within}} \in [522, 1303]$ MW.
- **Flat** $\mathcal{F} = \{1,2,3\}$: overnight; $\sigma_{\text{within}} \in [336, 357]$ MW.
- **Midday** $\mathcal{M} = \{11,\dots,14\}$ (solar-marginal, $\sigma_{\text{within}} \in [468, 502]$ MW): excluded from main specs; kept for falsification (Appendix A.2).

Descriptive critical − flat differential. Let $D^{\bullet}_{d,h}$ stand for any one of the five within-hour dispersion outcomes of §4.4 on date d, clock-hour h. Fit per outcome on the post-MTU15 sample, $h \in \mathcal{C} \cup \mathcal{F}$:

$$D^{\bullet}_{d,h} \;=\; \eta_d \;+\; \theta\,\mathbf{1}\{h \in \mathcal{C}\} \;+\; \varepsilon_{d,h}$$

with date fixed effects η_d (distinct from the bandwidth δ). θ is the critical−flat dispersion gap; a descriptive contrast, not a causal estimate.

Outcome	Subsample	ID15	DA15	crit/flat ratio (IDA, DA)
$D^{\bar{p}}$ (EUR/MWh)	per unit (pooled)	**0.39***** (0.05)	**1.35***** (0.11)	4.1×, **7.4**×
D^{m} (CV)	per unit (pooled)	**0.024***** (0.003)	**0.033***** (0.002)	4.3×, 4.9×
D^{MCP} (EUR/MWh)	system	**4.55***** (0.40)	**3.97***** (0.37)	2.8×, 2.6×
D^{σ_p} (EUR/MWh)	CCGT	−0.19 (0.36)	**0.61***** (0.12)	0.8×, **6.5**×
	Hydro	**0.48***** (0.08)	**1.32***** (0.17)	2.2×, 3.3×
	Hydro pump	**0.14***** (0.03)	**0.70***** (0.17)	3.3×, 10.9×
	Wind	0.00 (0.01)	0.03** (0.01)	0.9×, 1.9×
$D^{N_{\text{eff}}}$	CCGT	0.17 (0.16)	**0.17***** (0.02)	1.1×, **14.3**×
	Hydro	**0.05***** (0.01)	**0.19***** (0.03)	2.0×, 2.8×
	Hydro pump	**0.03***** (0.01)	**0.13***** (0.03)	2.6×, 4.3×
	Wind	0.00 (0.00)	0.01 (0.00)	1.0×, 1.3×

Table 3: Post-MTU15 critical−flat within-hour dispersion gap θ, by outcome. SE clustered by date; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (applies to all subsequent tables). Bid-level rows ($D^{\bar{p}}$, D^{m}, D^{MCP}) are bandwidth-free; the per-tech D^{σ_p} and $D^{N_{\text{eff}}}$ rows use the window-specific p_{90} bandwidth (§4.3): ID15 column δ=62 (post-ID15 IDA); DA15 column δ=50 (post-DA15 DA).

[3] Spanish-market convention calls this the *hueco térmico* (thermal gap) — the load that thermal and other dispatchable generation must cover after subtracting wind and solar.

5 Empirical strategy: three specifications for three economic objects

Indices follow §4.2. The Spec C treatment indicator is $\mathbf{1}\{d \geq T_r\} \cdot \mathbf{1}\{h(p) \in \mathcal{C}\}$ with the sample restricted to $h(p) \in \mathcal{C} \cup \mathcal{F}$.

5.1 Spec A: Bayesian Structural Time Series for prices, cleared energy, and wedge SD

Takeaway. Spec A is the Bayesian state-space counterfactual: train on a long pre-window with year-by-year wind and solar interactions, extrapolate forward to the post-window, read the treatment effect as the gap between observed and counterfactual. The year-by-year renewable interactions are the load-bearing element — Spanish wind+solar capacity grew ∼6× over 2018–2025, so a flat coefficient mis-fits the cross-year merit-order shift. A 2024 same-calendar placebo with a fake cutover guards against recurring seasonal artefacts.

Idea. A Bayesian Structural Time Series (BSTS) model (Brodersen et al., 2015), as applied to the EPEX 15-min reform by Märkle-Hußet al. (2020), fits a flexible state-space decomposition of the response on the pre-reform window and extrapolates forward to construct a counterfactual: what the outcome *would* have looked like in the post-window absent the reform. The treatment effect is the gap between observed and counterfactual. The strength over other methods is that BSTS handles complex seasonality and trend *internally*, useful here since the pre-windows are short.

Model. The daily response decomposes additively into trend, weekly cycle, exogenous covariates, and noise:

$$y_t \;=\; \underbrace{\mu_t}_{\text{trend}} \;+\; \underbrace{\gamma_t}_{\text{weekly cycle}} \;+\; \underbrace{X_t'\beta}_{\text{exogenous covariates}} \;+\; \underbrace{\varepsilon_t}_{\text{noise}}.$$

μ_t is a random walk with stochastic slope (a slowly evolving local level), γ_t a 7-day cycle, and X_t a covariate vector consisting of contemporaneous wind GWh, solar GWh, gas price (EUR/MWh), *and* the interactions $\text{wind}_t \times \mathbf{1}\{\text{year}(t) = y\}$ and $\text{solar}_t \times \mathbf{1}\{\text{year}(t) = y\}$ for each calendar year y in the pre-window (one calendar year omitted as reference). A spike-and-slab prior on β shrinks unused regressors. Demand is excluded to avoid simultaneity. The year-by-year renewable interactions are the load-bearing element of the spec: Spanish wind+solar installed capacity grew by a factor of ∼6 over 2018-2025, so the renewable price coefficient is not constant across the pre-window. Treating it as constant under-fits the cross-year merit-order shift and reports artificially confident treatment effects (Appendix A.1 sweeps the diagnostic). Demand is excluded to avoid simultaneity.

Outcomes y_t. DA or IDA daily-average clearing prices, per-tech daily cleared GWh, within-day wedge SD Δ_d^{SD} overall and per hour-class $\Delta_d^{\text{SD},\mathcal{X}} = \text{sd}_{h\in\mathcal{X}}(\Delta_{d,h})$ for $\mathcal{X} \in \{\mathcal{C}_M, \mathcal{C}_E, \mathcal{M}, \mathcal{F}\}$, and per-tech IDA in-band sell-share $\text{SS}_{t,d} = \sum q_{t,d}^{\text{sell, in-band}} / \sum(q_{t,d}^{\text{sell, in-band}} + q_{t,d}^{\text{buy, in-band}})$ (§6.3).

Estimand. Conditional on X_t, the post-cutover counterfactual equals the pre-trained extrapolation $\hat{y}_t(0)$. The average treatment effect on the post-window is

$$\bar{\tau} \;=\; \tfrac{1}{|\text{post}|} \textstyle\sum_{t \geq T_r} \big(y_t - \hat{y}_t(0)\big),$$

reported together with its 95% credible interval from the MCMC posterior produced by the `CausalImpact` R package.

Windows. Pre-windows are long: 2022-01-01 onwards through the day before each reform cutover. The long pre-window is what makes the year-by-year renewable interaction identified (each calendar year contributes its own renewable response). Post-windows are short: 40 days at each cutover. For ID15 the 40-day post-window stops at the April 2025 blackout; for DA15 the pre-window straddles the blackout so the post-window does not need a separate reforzada-constant truncation. (A pre-window that respects the regulatory regime in a narrower sense — short-pre BSTS with a flat renewable coefficient — is reported as a robustness check in Appendix A.1; the headline spec uses the long pre with year interactions because the short-pre flat-coefficient configuration mis-fits the cross-year renewable shift and reports artificially confident effects on multiple level outcomes.)

Placebo cross-check. The same BSTS is re-run on a same-calendar window one year before each reform with a fake cutover. The real-window 95% credible interval must exclude zero AND the placebo must be either null or opposite-signed for a finding to be retained as a reform-attributable effect; placebos that move in the same direction as the real estimate flag a recurring calendar shift the seasonality model has not absorbed.

MCMC and prior settings. `niter` = 2,000 MCMC iterations after burn-in (a 5× increase changes point estimates and 95% intervals by $< 2\%$ across all reported cells, Appendix A.1). `prior.level.sd` = 0.005 as a fraction of the response SD; the result is insensitive to this choice over $[0.005, 0.01]$ but distorts under an aggressive 0.001 (Appendix A.1).

5.2 Spec B: per-hour-class BSTS on in-band offered energy

Takeaway. Spec B is Spec A applied per (tech, market, hour-class) cell with the in-band offered-energy outcome. It lets the data reveal any critical-hour concentration as heterogeneity across hour classes rather than imposing the critical/flat differential as the identifying restriction. Under the year-interaction control no Spec B cell survives as a retained finding; the bid-shape DiD of Spec C provides the cleaner within-day identification on the same channel.

5.3 Spec C: per-curve DiD for bid shape

Takeaway. Spec C identifies WITHIN-DAY changes in the bid curve: per (unit, date, period) we summarise the in-band ladder with four jointly informative statistics (intercept α, slope β, curvature γ, effective tranche count N_{eff}) and run a DiD that uses flat hours as the within-day control for critical hours. The flat-hour partition is the identifying restriction; no time-series extrapolation is needed.

For per-tech bid-shape outcomes $Y \in \{\hat{\alpha}, \hat{\beta}, \hat{\gamma}, N_{\text{eff}}\}$ (with σ_p kept as the legacy diagnostic; see §4.3) we use the DiD:

$$Y_{u,d,p[,s]} = \alpha_u + \beta\, \mathbf{1}\{d \geq T_r\} + \xi\, \mathbf{1}\{h \in \mathcal{C}\} + \theta\, D_{d,h} + \varepsilon, \qquad \text{SEs clustered by date.}$$

where $\text{slope}_{u,d,p}$ and $\text{intercept}_{u,d,p}$ are jointly the OLS coefficients of bid price on cumulative in-band MW within the curve. Sort the in-band tranches by price (re-indexing $k = 1, \ldots, |\mathcal{B}_{u,d,p}|$ over the in-band set $\mathcal{B}_{u,d,p}$ defined in §4.3); let $Q_k = \sum_{j \leq k,\, j \in \mathcal{B}_{u,d,p}} \Delta q_j$ be the cumulative MW after the kth in-band tranche; the OLS fit on the in-band step endpoints is

$$p_k \;=\; \alpha \,+\, \beta\, Q_k \,+\, e_k, \qquad k = 1, \ldots, |\mathcal{B}_{u,d,p}|,$$

run unweighted (each tranche contributes one data point with equal weight; no MW-weighting). The slope β is supply-curve steepness in the competing tier (EUR/MWh per MW) — the most direct empirical counterpart of the model's per-product residual-demand slope b_{mt}. The intercept α is the predicted price at $Q = 0$, the y-intercept of the OLS line through the in-band step endpoints; geometrically it measures the level of the in-band ladder (where the line sits on the price axis), and pairs naturally with the slope to describe how the bid curve translates and rotates across regimes.

The four outcomes are non-redundant: $\hat{\alpha}$ pins the level of the ladder at zero cumulative MW, $\hat{\beta}$ captures the rate at which price climbs per MW, $\hat{\gamma}$ captures the curvature (convex or concave bend), and N_{eff} summarises the discrete tranche-count density. A bid curve can be a single block (low slope, high $\hat{\alpha}$, $N_{\text{eff}} = 1$), a sparse steep staircase (high slope, $\hat{\gamma}$ near zero, low N_{eff}), a dense narrow ladder (low slope, high N_{eff}, $\hat{\gamma}$ near zero), or a convex curve (low slope at low MW, steep at high MW, $\hat{\gamma} > 0$); reading all four jointly disentangles level from slope from curvature from density. The justification is mechanism-based. **The granularity reform's bid-shape effect is structurally a critical-hours effect.** Differentiated bidding only pays off where the residual demand curve is steep enough to reward it: in critical hours that condition holds, so 4 quarter-prices instead of 1 unlock real price discrimination; in flat hours the curve is shallow and the extra granularity is mostly wasted optionality. The (critical − flat) differential θ therefore IS the granularity-enabled discrimination, not a residual we are netting out. Flat hours act as a low-exposure within-day control — not a zero-exposure one, since MTU15 reaches them too.

For the level outcomes — prices and cleared MW in §5.1, in-band offered energy per hour-class in §5.2 — we keep BSTS: per-hour-class BSTS already lets the data reveal any critical-hour concentration as heterogeneity across the four estimates, without imposing the critical-flat differential as the identifying restriction.

Since each MTU15 reform is studied separately, this is a standard 2×2 DiD without staggered-treatment negative weights (de Chaisemartin and D'Haultfœ uille, 2026, ch. 3).

6 Results

We report four empirically identified channels and one descriptive system-cost finding. **Identification**: Spec C identifies bid-shape effects within day via the critical/flat contrast; Spec A identifies level outcomes via long-pre BSTS with year-by-year renewable interactions. The model predicts several other level shifts (granular-side price drop, Cournot scale-up of cleared MWh, in-band offered-energy migration) whose point estimates have the right sign but whose credibility intervals do not exclude zero under the year-interaction Spec A; we report those in §6.5 as non-retained and present the diagnostic sweep in Appendix A.1.

6.1 Bid-shape widening in critical hours (Spec C)

Takeaway. The granularity reforms reshape the in-band bid curve in critical hours. DA15 combined-cycle gas day-ahead translates up ($\hat{\alpha} \approx +28$ EUR/MWh), flattens ($\hat{\beta} \approx -0.15$ EUR/MWh per MW), and bends convex ($\hat{\gamma} > 0$) — Cournot scale-up. ID15 day-ahead intraday-cleared CCGT (anticipation) translates down ($\hat{\alpha} \approx -40$), steepens ($\hat{\beta} \approx +0.27$), bends concave ($\hat{\gamma} < 0$). ID15 intraday-auction hydro and pump-storage add tranche density without changing level or slope. Demand-side buyers in critical hours raise their willingness-to-pay at the top of the ladder by EUR +15/MWh on the day-ahead leg and EUR +33/MWh on the intraday leg, mirroring the sell-side level shift and consistent with equilibrium re-coordination rather than a one-sided supply move.

Outcome (DiD θ)	Tech	ID15 DA	ID15 IDA	DA15 DA	DA15 IDA
$\hat{\alpha}$ (EUR/MWh)	CCGT	$\mathbf{-40.2}^{***}$	11.8^{*}	$\mathbf{28.3}^{***}$	6.0^{**}
	Hydro	0.2	87.3	19.7	$\mathbf{9.3}^{***}$
	Pump-storage	1.1	−3.0	−1.8	16.1
$\hat{\beta}$ (EUR/MWh per MW)	CCGT	$\mathbf{0.27}^{***}$	-0.88^{*}	$\mathbf{-0.15}^{***}$	0.20
	Hydro	0.15	−1.50	$\mathbf{-1.79}^{***}$	−0.01
	Pump-storage	−0.01	0.07	0.01	−0.25
$\hat{\gamma}$ ($\times 10^3$)	CCGT	$\mathbf{-3.99}^{***}$	10.0	$\mathbf{1.03}^{***}$	7.6
	Hydro	−305	−109	$\mathbf{-7{,}494}^{***}$	1.6
	Pump-storage	0.07	1.45	0.05	−0.16
N_{eff}	CCGT	−0.08	0.79	$\mathbf{1.89}^{***}$	−0.17
	Hydro	-0.24^{***}	$\mathbf{0.18}^{***}$	$\mathbf{0.51}^{***}$	0.00
	Pump-storage	0.07	$\mathbf{0.12}^{*}$	−0.06	−0.16

Table 4: Spec C per-curve DiD on bid-shape outcomes: intercept $\hat{\alpha}$, linear slope $\hat{\beta}$, curvature $\hat{\gamma}$, and effective tranche count N_{eff}. SEs clustered by date.

Mechanism (joint $\hat{\alpha} + \hat{\beta} + \hat{\gamma}$). The geometric reading of the bid curve sharpens once curvature is included.

DA15 DA CCGT: $\hat{\alpha}\uparrow$, $\hat{\beta}\downarrow$, $\hat{\gamma}\uparrow$ (convex). The curve translates up, has a flatter average slope, and bends *convex* $\Rightarrow$ locally flat at low MW (more in-band capacity at competitive prices — Cournot scale-up at (Q, Q), mechanism (iv) of §3.5) and locally steep at high MW (forward-side ladder widening, mechanism (iii)). The CCGT signature is robust to wider bandwidths ($\hat{\beta}$ steepens monotonically from −0.15 at p_{90} to −0.49 at $\delta = 200$; $\hat{\gamma}$ stays convex at +0.001 to +0.006, see Table 11); a small same-sign pre-trend (Table 7) means the reform-period coefficient is continuation-plus-augmentation rather than a clean jump. Hydro shows the same headline joint signature on the p_{90} window, but its $\hat{\gamma}$ collapses from −7.6 at p_{90} to near zero at $\delta = 200$ — hydro's apparent strong curvature is a narrow-band artefact, not a robust shape change.

ID15 DA CCGT (anticipation): $\hat{\alpha}\downarrow$, $\hat{\beta}\uparrow$, $\hat{\gamma}\downarrow$ (concave). The curve translates down, steepens on average, bends *concave* $\Rightarrow$ aggressive climb at low MW with a flatter tail at high MW: pure price discrimination on the still-hourly DA leg, no scale-up. The opposite curvature sign at the two reforms for the same (tech, market) cell is direct evidence that the same firm responds with two distinct mechanisms (Stage-1 option-value channel, §3.5(v)).

DA15 DA Pump-storage: not retained. The earlier 40-day post-window read $\hat{\alpha}\downarrow$, $\hat{\beta}\uparrow$ as a storage-arbitrage instrument, but extending the post-window to 148 days (the full 2026-02-26 OMIE coverage) collapses both coefficients toward zero ($\hat{\alpha} = -1.7$, $\hat{\beta} = 0.01$, both insignificant). The original sign was a short-window artefact; we no longer claim a DA15 pump-storage bid-shape effect.

ID15 IDA hydro/pump-storage: $N_{\text{eff}}\uparrow$ with no level/slope shift $\Rightarrow$ added tranche density on the spot leg (mechanism M4 of §3.4). The three retained joint-signature stories are visualised in Figure 5.

6.2 Production-unit arbitrageurs are empirically active in intraday auctions

Takeaway. 4–30% of intraday-auction unit-sessions carry both a sell and a buy offer, with the share ranging from 20% for nuclear to 83% for wind. The sequential-market theory's arbitrageur

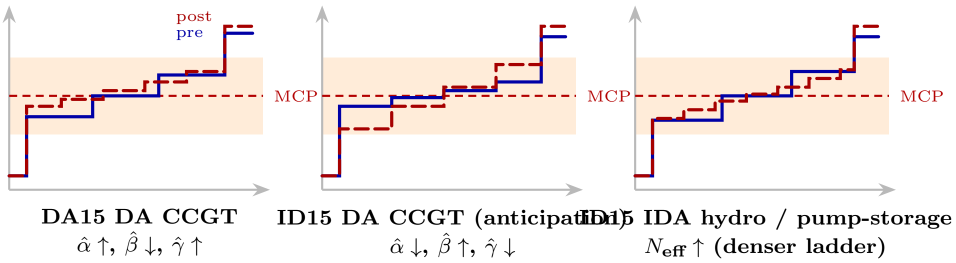
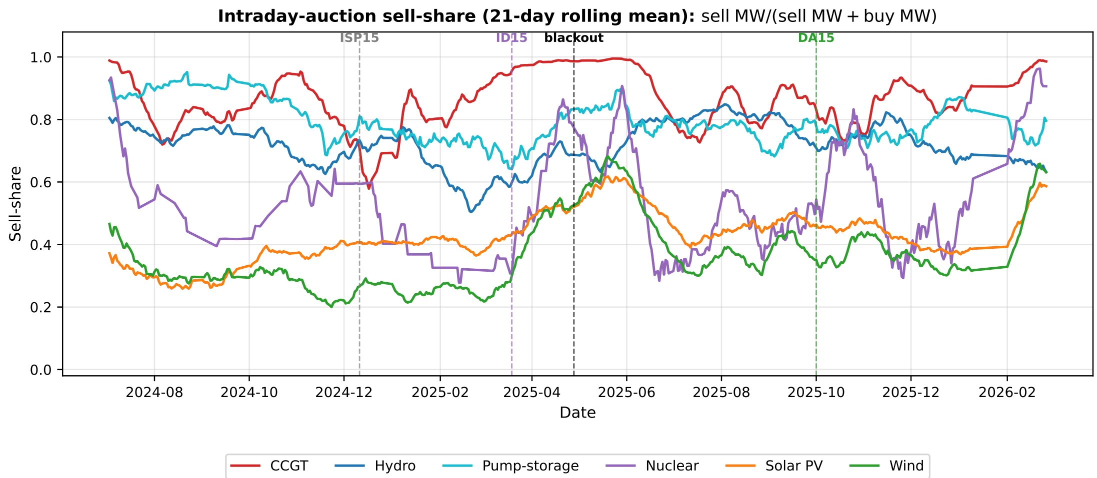


Figure 5: Three retained stylised bidding shifts under the 154-day DA15 post-window. Blue solid: pre-reform; red dashed: post-reform. The original draft contained a fourth DA15 pump-storage panel that was a 40-day window artefact (§6.1); it is dropped.

agent is observable on the bid stack across every production technology.

Production units submit sell offers in both markets but buy offers only in IDA (the single-product-per-hour DA admits no within-auction spread). Figure 6 plots the daily IDA sell-share per tech; the %-dual mini-table shows production-unit arbitrageurs are active across every tech (4–30% of unit-sessions carry both sides).



	CCGT	Hydro	Pump-storage	Nuclear	Solar PV	Wind
IDA \% dual sessions	30.6\%	43.3\%	36.1\%	19.7\%	58.8\%	82.7\%

Figure 6: IDA sell-share by tech, 21-day rolling mean. Vertical dashes: ISP15, ID15, blackout, DA15. %-dual table below reports the share of unit-sessions with both sell and buy.

The IDA %-dual row in Figure 6 varies from 20% (nuclear) to 83% (wind), with renewables and pump-storage well above 35%. This confirms an arbitrageur agent of the kind the sequential-market theory presupposes is empirically operating on the intraday bid stack across every production technology actually clearing volume — the empirical confirmation of P7 (§3.6).

6.3 IDA in-band sell-share rises at ID15 (Spec A)

Takeaway. The intraday auction stops being a pure rebalancing venue and becomes a per-quarter pricing instrument: nuclear +39pp, wind +26pp, pump-storage +13pp on the share of in-band activity that is sell rather than buy. No 2024 placebo because the pre-IDA-reform sell-share was a constant 1.0.

ID15 lifts the daily IDA in-band sell-share sharply for nuclear (38.8pp), wind (26.0pp) and pump-storage (13.0pp); see Table 5. Same-calendar 2024 placebo unavailable (pre-IDA-reform sell-share constant at 1.0); falsification by descriptive break in Figure 6.

Mechanism. Pre-ID15 the IDA was a rebalancing venue; ID15 turned it into a per-quarter pricing instrument, so units shifted activity onto the sell side to capture within-hour rent (mechanism M3 of §3.4).

6.4 In-band offered energy migration (Spec B): not retained

Takeaway. Spec B's per-hour-class in-band offered-energy cells (24 cells across tech × market × hour-class) deliver model-consistent signs but credible intervals that bracket zero under the year-by-year renewable-interaction control. The bid-shape DiD of §6.1 identifies the same Cournot-scale-up channel through a within-day critical/flat control; we drop Spec B as a primary finding.

6.5 Level-shift predictions that the data cannot identify

Takeaway. The model's four level-shift predictions (granular-side price drop, Cournot cleared-MWh scale-up, in-band offered-energy migration, renewable cleared-MW reallocation) deliver right-sign point estimates but 95% credible intervals that bracket zero once the renewable coefficient is allowed to vary across calendar years. We drop them as separate findings and read the same economic content through the within-day bid-shape DiD of §6.1.

Spec A applies BSTS to system-level outcomes: clearing prices, per-tech daily cleared GWh, and the within-day wedge SD Δ_d^{SD}. Table 5 reports the Real, Placebo, and placebo-net (Net) posterior means for each outcome and each reform.

Outcome	**ID15 real**	**2024 placebo**
Within-hour DA−IDA wedge SD (EUR/MWh)		
Overall	**1.15****	−0.96
Critical hours	**2.12*****	−0.68
IDA in-band sell-share (pp)		
Nuclear	**38.8*****	n/a
Wind	**26.0*****	n/a
Pump-storage	**13.0*****	n/a

Table 5: Retained Spec A findings (long-pre BSTS with year-by-year renewable interactions). Full table in Appendix A.1.

6.6 Within-hour wedge SD jumps under ID15 and persists at DA15 (Spec A)

Takeaway. The day-ahead/intraday wedge MEAN stays flat across regimes (the Ito-Reguant scarcity wedge); the wedge SD jumps at ID15 (+1.15 EUR/MWh overall, +2.12 in critical hours) and does NOT collapse back at DA15 — the joint signature of M3 limited-arbitrage at $(1, Q)$ and (ii) per-product limited arbitrage at (Q, Q). The reform reshapes the second moment, not the mean.

Wedge mean is positive and stable across regimes (1 to 2 EUR/MWh) — the Ito-Reguant scarcity wedge $p_1 > p_2 > c$ under limited arbitrage. Reform-attributable change in the level is null; the variation is in the second moment.

Mechanism. ID15: 15-min IDA against 60-min DA ⇒ four IDA quarter-prices spread across each clock-hour vs one DA price, mechanically generating a within-hour DA−IDA gap with SD that tracks within-hour residual-demand variation. DA15: per-product arbitrage capacity binds at K/Q; residual wedge SD stays at $\sigma_\mu/(4b_1)$ instead of collapsing to zero (mechanism (ii) of

§3.5). ISP15 alone (settlement-only) produces a same-sign jump ⇒ bidders respond to the finer settlement reference itself.

6.7 Aggregator vs strategic-bidder footprint by reform

Takeaway. The two reforms reshape behaviour through different agents: ID15 surfaces the aggregator-side response (IDA sell-share rise on the spot leg) and a within-hour wedge SD that does not collapse; DA15 surfaces the strategic-bidder response (CCGT day-ahead ladder widening on the forward leg). The model's two-agent typology delivers a one-to-one match with the observed reform-by-reform incidence.

6.8 System-cost decomposition: granularity shrinks ajuste; reforzada inflates Fase I

Takeaway. Two policies running on the same calendar window pull system cost in opposite directions: the granularity reforms shrink the up-direction adjustment channels REE calls (real-time redispatch −1/3, aFRR up −1/2) by absorbing within-hour variation into the auction; the post-blackout reforzada regime separately inflates Fase I up-redispatch (+5M EUR/day, almost entirely CCGT voltage support). The net headline rise is reforzada-driven, not granularity-driven.

Within the pre-blackout zone (ISP15-win → MTU15-IDA pre-blk): total per-day ajuste cost drops from ∼EUR 16M to ∼EUR 14M; real-time redispatch falls by ∼1/3, aFRR up by ∼1/2, down-direction channels shrink in parallel. Quarter-level cleared schedules absorb within-hour variation that previously had to be cleaned up ex post.

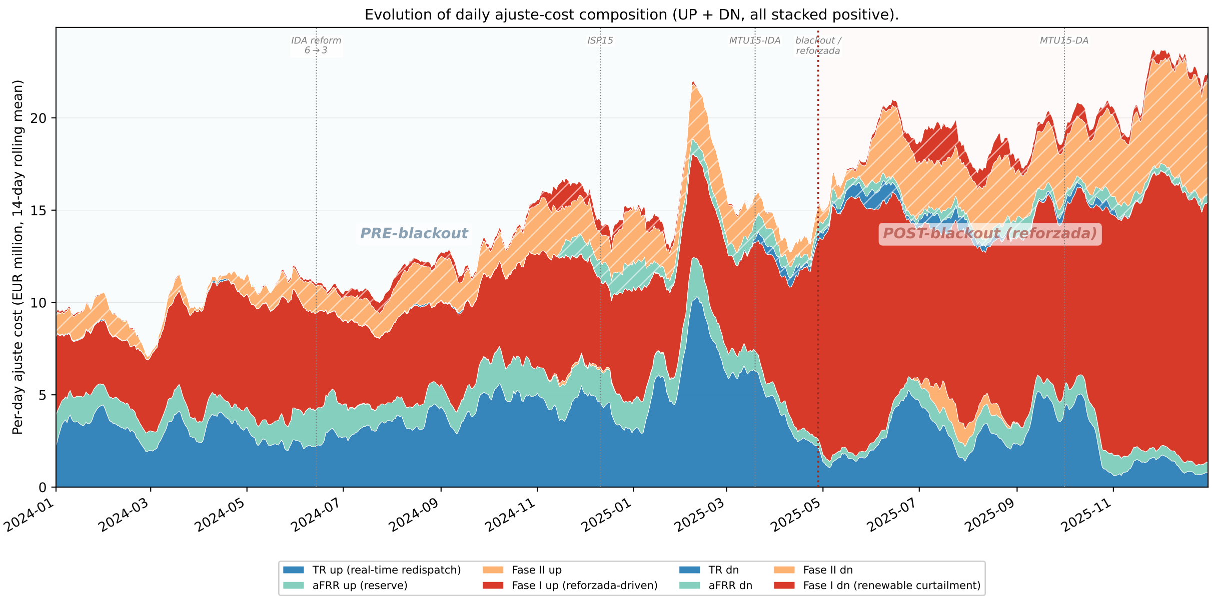


Figure 7: Daily ajuste-cost composition (EUR M / day, 14-day rolling mean): UP solid, DN hatched. Blue: pre-blackout. Red: post-blackout reforzada.

Post-blackout: Fase I up balloons by ∼5M/day (CCGT voltage support) and Fase II dn by ∼2–3M/day. These are reforzada-driven, not MTU15. The granularity channels keep shrinking inside the post-blackout zone but are overwhelmed by reforzada growth in the headline total. The post-blackout spike is CCGT-only (Appendix C.3) — a voltage-support intervention, not a granularity effect.

7 Conclusion: granularity is a strategic-latitude reveal, not a market-power amplifier

Headline. A market-microstructure reform that increases trading-period resolution acts as a *strategic-latitude reveal*, not a technology shock. Marginal costs, capacity, and the set of participants are unchanged; what changes is who has room to discriminate within the hour and where. The equilibrium response traces exactly that — combined-cycle gas restructuring its day-ahead ladder in critical hours, intraday hydro and pump-storage adding tranche density, demand-side buyers raising their reservation prices in step with sell-side level shifts, and operational aggregators using the finer grid to manage imbalance.

Does granularity amplify market power? No. The day-ahead/intraday wedge mean is positive at EUR 1–2/MWh and stable across regimes; none of the reforms shifts the level. The strategic engagement we document is firms competing more granularly, not extracting rents. The wholesale clearing-price level is where identification is weakest: a ~ -33 EUR/MWh ID15 drop is visible in both auctions but does not separate cleanly from forecast noise once the renewable coefficient is allowed to vary across years.

Does it harm consumers? Net of the post-blackout regulatory regime, no. The granularity reforms shrink the up-direction adjustment-cost channels REE has to call — real-time redispatch and aFRR[4] both fall across the reform sequence (Figure 7). The only adjustment channel that grows over the same window is Fase I[5] up-redispatch, and that growth is driven by the post-blackout reinforced-operation regime, not by MTU15. The two policies overlap calendar-wise but operate on different mechanisms (Appendix C.3).

Is ID15 implicated in the April 2025 blackout? The evidence does not support a direct attribution but sharpens a question worth flagging. Three facts: (i) REE's reinforced-operation response is technology-specific — combined-cycle gas dispatch roughly doubles after the blackout while solar and hydro are curtailed, consistent with a voltage-support stance rather than a market-driven outcome; (ii) spring renewable supply was nearly identical in 2024 and 2025 (solar 153 vs 143 GWh/day, wind 163 vs 146 GWh/day in the three weeks around the date), yet only 2025 blacked out, so the cleanest systemic change between the two springs is the three-step MTU15 sequence; (iii) over the 40 days between ID15 and the blackout, nuclear auction-plus-bilateral output dropped by -143 GWh/day net of the 2024 same-calendar placebo — significant at 1% and absent in the post-blackout DA15 window (Appendix A.8). The cleanest narrative is a chain — ID15 made bidding more aggressive on the granular leg, day-ahead prices ran near EUR 20/MWh with several near-zero days, and inflexible nuclear baseload (which carries the bulk of synchronous inertia) had limited room to operate — but each step is testable and none alone identifies the blackout's causal trigger. The ENTSO-E Expert Panel's final report (ENTSO-E Expert Panel, 2026) does not list ID15 as a contributing factor; its root-cause tree points to voltage- and reactive-power-control failures rather than to market microstructure.[6]

Anecdotally, the parallel CNMC report does flag the move to quarter-hourly trading as one

[4] aFRR (automatic Frequency Restoration Reserve): units committed to an aFRR band are automatically signalled to ramp within seconds to correct system frequency deviations.

[5] Fase I: REE's primary up-redispatch market; pay-as-bid restriction offers procured after OMIE clears.

[6] ENTSO-E technical root causes: renewable plants at fixed power factor, manually-operated shunt reactors, divergent overvoltage protection settings, converter-driven instability, and absence of power-system stabilisers on several large units.

contributor to the abrupt voltage variations observed in the months around the incident (CNMC, 2026, p. 44):

> *It is worth recalling that the implementation of the 15-minute trading unit in the Spanish intraday market was completed on 18 March 2025, one month before the incident, with the entry into force of quarter-hourly trading in both the intraday auctions and the continuous intraday market. The implementation was carried out within the framework of the European intraday coupling projects in line with Regulation (EU) 2017/2195. The evolution to quarter-hourly trading and settlement has made the appearance of transitions between negative and positive prices increasingly frequent, leading to strong changes in production volumes in certain zones.*[7]

Answering whether market-microstructure reforms contributed to the macro-conditions that made the proximate triggers cascade — frequent within-hour swings, converter-driven volatility under fixed-power-factor renewable dispatch — is beyond this thesis's scope and requires electrical-engineering knowledge we do not bring. The CNMC's flag points exactly at that gap.

[7] Original (CNMC report, p. 44): *"A este respecto, cabe recordar que el proceso de implantación de la unidad de negociación de 15 minutos en el mercado intradiario español se completó el 18 de marzo de 2025, 1 mes antes del incidente, con la entrada en vigor de la negociación cuarto-horaria tanto en las subastas intradiarias como en el mercado intradiario continuo. La implementación se realizó en el marco de los proyectos europeos de acoplamiento intradiario en coherencia con el Reglamento (UE) 2017/2195. La evolución a una negociación y liquidación de productos cuarto horarios ha hecho cada vez más frecuente la aparición de transiciones entre precios negativos y positivos que conducen a fuertes cambios del volumen de producción en determinadas zonas."*

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A Robustness

A few caveats up front: bid-shape DiDs in a market this complex (overlapping reforms, regulatory regime changes, weather-driven dispatch) will not deliver textbook causal cleanliness on every cell. The robustness checks below flag where the headline reading is solid and where it rests on shakier ground.

A.1 Year-by-year solar coefficient: the renewable response is non-stationary

The Spec A BSTS uses long pre-windows (2022 onwards) with $\text{wind}_t \times \mathbf{1}\{\text{year} = y\}$ and $\text{solar}_t \times \mathbf{1}\{\text{year} = y\}$ interactions. The economic reason is that Spanish wind+solar installed capacity grew by a factor of ~ 6 over 2018-2025, and the marginal effect of an additional GWh of renewable output on price is not constant across that period. A flat-coefficient spec mis-fits the cross-year merit-order shift and reports artificially confident treatment effects (§5.1).

Table 6 reports the cumulative solar coefficient (base + year-dummy) by calendar year, extracted from the BSTS posterior of the price spec under both windows (ID15 pre ends 2025-03-18, DA15 pre ends 2025-09-30). The spike-and-slab inclusion probability for the wind×year interactions is essentially zero (always below 1%); the wind coefficient is stable around -0.50 EUR/MWh per GWh. The solar coefficient is the one that varies.

Year	ID15 IDA spec	DA15 IDA spec	DA15 DA spec
2022 (base)	−0.22	−0.23	−0.22
2023	−0.02	−0.23	−0.17
2024	−0.46	−0.47	−0.40
2025 †	**−0.29** *(Jan-Mar only)*	**−0.50**	**−0.49**

Table 6: Cumulative solar coefficient (EUR/MWh per GWh of solar) by year, extracted from the BSTS posterior of the corresponding price spec. Cumulative coefficient = base + year-dummy interaction (conditional posterior mean among MCMC draws that include the regressor). †The ID15 spec pre-window ends 2025-03-18, so the 2025 × solar coefficient is fitted on Jan-Mar 2025 only, when Spanish solar production is at its lowest seasonal level; the DA15 spec pre-window extends through 2025-09-30 and includes summer, giving the more representative estimate.

Reading the table: the DA15 spec (which sees Jan-Sep 2025) tells the cleaner story. Solar's marginal price-suppressing effect is roughly constant from 2022 to 2023, then nearly doubles from 2023 to 2024 (Spain added $\sim$5 GW of solar capacity in 2024 alone), and continues to deepen in 2025 as solar saturates more clock-hours and negative-price events become routine. The ID15 spec's 2025 coefficient looks deceptively small (Jan-Mar winter sun is weak) but the underlying renewable response in 2025 is the most negative on record. The non-stationarity is exactly what the year interaction is designed to absorb: a flat-coefficient spec would average -0.22 (2022) and -0.50 (2025) into a single value around -0.35, mis-fitting both ends of the window.

This diagnostic also clarifies why several level findings (§6.5) lose significance under the year-interaction spec: forecast uncertainty about what solar's marginal price impact will be in the post-cutover window is large precisely because the coefficient has been shifting fast in the years immediately preceding the cutover.

Note on bid-shape outcome in this appendix. The Spec C bid-shape robustness checks below were originally specified on σ_p (the MW-weighted in-band price SD), which was a headline outcome in earlier drafts. The main text now uses the quadruple $(\hat{\alpha}, \hat{\beta}, \hat{\gamma}, N_{\text{eff}})$ (§4.3), with σ_p

relegated to a legacy diagnostic because it is mechanically dominated by $(\hat{\beta}, \hat{\gamma})$: for a near-linear curve $\sigma_p \approx |\hat{\beta}| \cdot \sigma_Q$, so the sign of a σ_p DiD coefficient is the sign of $|\hat{\beta}|$ scaled by the in-band MW spread, and curvature-driven dispersion shows up symmetrically in $\hat{\gamma}$. The robustness signs and significance reported below on σ_p therefore carry over qualitatively to $\hat{\beta}$ (and via curvature also to $\hat{\gamma}$). Per-cell re-runs of the placebo, hour-class falsification, cross-border and per-firm checks on the new outcomes $(\hat{\alpha}, \hat{\beta}, \hat{\gamma})$ are scheduled for the final submission.

A.2 Pre-trend placebo — the headline check

A pre-only midpoint placebo splits each reform's pre-window at its midpoint and runs the same critical-flat DiD on each bid-shape outcome (Table 7). Genuine reform effects should leave the placebo near zero or running with a sign-opposite or magnitude-smaller pattern than the headline. Figure 8 plots the weekly CCGT in-band step series for critical and flat hours on both markets as the visual analogue.

The ID15 own-market cells survive the placebo cleanly: CCGT α collapses from -40.2 (headline) to $+1.8$ (placebo) and γ from -0.0040 to $+0.0015$ — sign flips, not magnitude reductions. The DA15 own-market CCGT cells are the headline find under stress: the placebo runs same-sign and same-order-of-magnitude as the real (α: $+28.3$ real, $+40.2$ placebo; γ: $+0.0010$ real, $+0.0022$ placebo). The real Cournot-style convex widening is therefore real *on top of* a pre-existing CCGT widening trend, not separable from it — we read the DA15 CCGT convexity as continuation-plus-augmentation rather than a clean reform jump, and the cross-section of the headline tells a stronger DA15 story because hydro and pump-storage show clean separation (hydro β real -1.79 vs placebo $+1.18$; pump-storage α real -1.8 insignificant vs placebo -4.3).

	ID15 IDA (own)		ID15 DA (cross)		DA15 DA (own)		DA15 IDA (cross)	
Tech	Hdln	Plac	Hdln	Plac	Hdln	Plac	Hdln	Plac
Intercept α (EUR/MWh; positive = higher curve)								
CCGT	11.8*	4.9	−40.2***	1.8	28.3***	40.2***	6.0**	3.8**
Hydro	87.3	10.2***	0.2	−0.4	19.7	53.3*	9.3***	6.3***
Pump-storage	−3.0	−12.7***	1.1	1.3	−1.8	−4.3	16.1	−4.9
Slope β (EUR/MWh per MW; more negative = steeper)								
CCGT	−0.881*	0.146	0.269***	0.004	−0.150***	−0.206***	0.200	−0.101
Hydro	−1.498	−0.156**	0.154	−1.703***	−1.785***	1.176***	−0.012	−0.022
Pump-storage	0.071	0.248	−0.010	0.015*	0.008	0.046***	−0.254	0.007
Curvature γ (EUR/MWh per MW^2; positive = convex)								
CCGT	0.0100	0.0334*	−0.0040***	0.0015***	0.0010***	0.0022***	0.0076	0.0002
Hydro	−0.1088	0.2455	−0.3055	−1.3947	−7.4938***	1.6198	0.0016	−0.0105
Pump-storage	0.0014	0.0005	0.0001	0.0002	0.0001	0.0000	−0.0002	−0.0002

Table 7: Pre-only midpoint placebo for Spec C bid-shape DiD, all four (reform, market) cells. "Hdln" reproduces the headline DiD from §6.1; "Plac" fits the same critical/flat DiD on the pre-window split at its midpoint. Stars: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.3 Hour-class falsification (midday vs. flat)

Re-running Spec C with midday hours (§4.5) substituted for critical hours covers all four (reform, market) cells (Table 8). DA15 own-market CCGT reshapes the entire non-flat zone — α, β and γ all carry the same widening sign in midday as in critical, magnitudes of comparable order; the reform did not engage critical hours alone. ID15 own-market CCGT, by contrast, has α

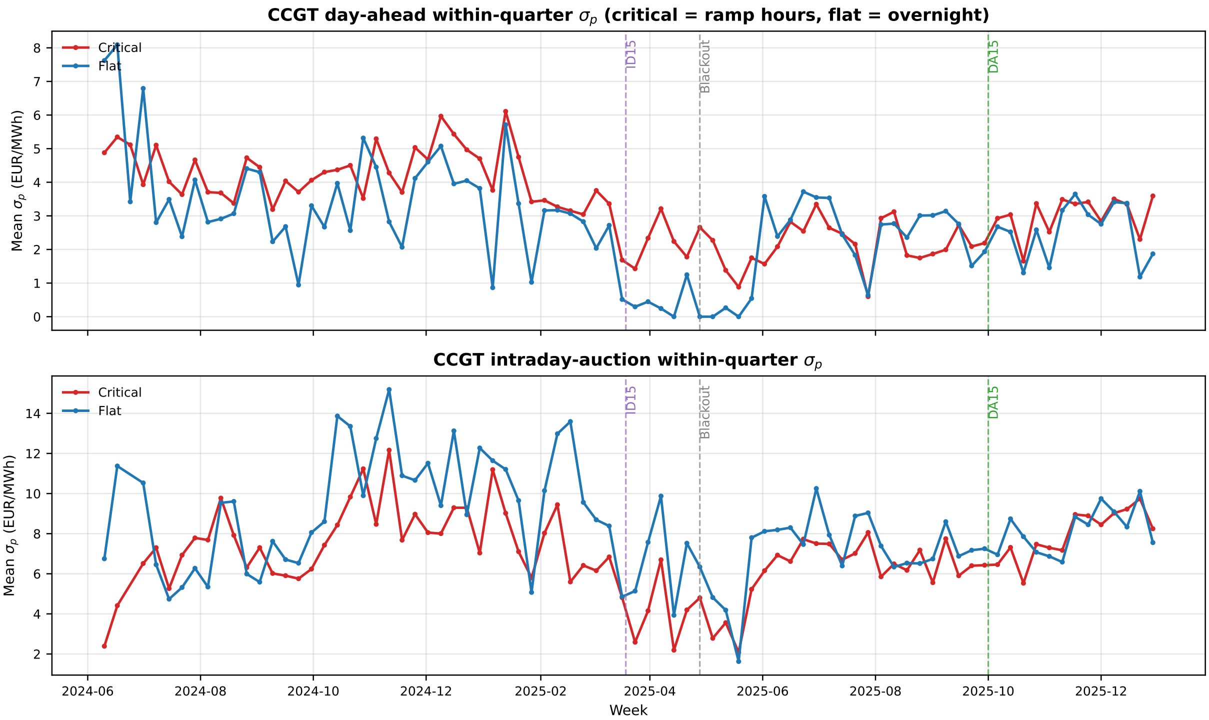

Figure 8: Weekly CCGT within-quarter σ_p by hour-class, 2024-06 to 2025-12. *Top:* day-ahead; *Bottom:* intraday-auction (latest-session in-band tranches, $|p_k - \text{MCP}| \leq 50$ EUR/MWh). Red: critical hours; blue: flat hours. Vertical dashes: ID15 (2025-03-19), blackout (2025-04-28), DA15 (2025-10-01).

moving in the opposite direction in midday relative to the (marginal) headline, consistent with combined-cycle gas not engaging the granularity instrument in the solar-saturated midday window when its expected acceptance probability is essentially zero. Hydro flips in places (the DA15 DA γ is concave in critical hours and convex in midday) — hydro's strategic re-shaping engages directionally with the hour-class structure, not symmetrically across the day.

	ID15 IDA (own)		**ID15 DA (cross)**		**DA15 DA (own)**		**DA15 IDA (cross)**	
Tech	Crit	Midd	Crit	Midd	Crit	Midd	Crit	Midd
Intercept α								
CCGT	11.8*	8.3	−40.2***	—	28.3***	1.2	6.0**	0.8
Hydro	87.3	84.3	0.2	14.8***	19.7	33.2***	9.3***	17.9***
Pump-storage	−3.0	−7.5	1.1	−6.3**	−1.8	21.3***	16.1	139.6
Slope β								
CCGT	−0.881*	−0.460	0.269***	—	−0.150***	−0.006	0.200	−0.013
Hydro	−1.498	−1.977	0.154	−2.306***	−1.785***	−4.025***	−0.012	−0.186
Pump-storage	0.071	0.146	−0.010	−0.023*	0.008	0.028***	−0.254	−1.508
Curvature γ								
CCGT	0.0100	−0.0002	−0.0040***	—	0.0010***	0.0006**	0.0076	0.0198*
Hydro	−0.1088	−0.0367	−0.3055	18.7***	−7.4938***	−34.1***	0.0016	−0.0079
Pump-storage	0.0014	0.0006	0.0001	−0.0000	0.0001	0.0002*	−0.0002	−0.0008

Table 8: Midday-vs-flat falsification, Spec C bid-shape DiD. Same construction as the headline of §6.1 with midday $\mathcal{M} = \{11, \ldots, 14\}$ substituted for critical hours $\mathcal{C} = \{5\text{–}8, 16\text{–}22\}$. A clean "—" indicates an unidentified cell (insufficient observations after restriction).

A.4 Cross-border flow control (ID15 IDA only by design)

The XBID continuous market and the SIDC intraday auctions both moved to 15-min on the same date as ID15 — a sibling treatment on the intraday side rather than an omitted-variable bias. Adding hourly Spain-France cross-border net flow as a control on the ID15 IDA σ_p DiD leaves the per-tech coefficients essentially unchanged (Table 9), confirming that the bid-shape widening is not absorbed by the contemporaneous cross-border-trade adjustment. We do not run the same control on the DA-side cells (ID15 DA cross-market and DA15 DA own-market), because XBID coupling is an intraday phenomenon and the day-ahead cross-border channel goes through SIDC implicit-allocation capacity that is fixed at the noon DA gate — a different variable conceptually and one that does not change at the same reform dates. The reforzada regulatory stance is the other natural confound; it acts on the *level* of Fase I up-redispatch (Appendix C.3) rather than on the within-quarter bid-shape outcomes used in Spec C, and the granularity-vs-reforzada separation is documented at the system level in §6.8.

	σ_p **DiD**		N_{eff} **DiD**	
Tech	no XB control	with XB control	no XB control	with XB control
CCGT	0.10	0.11	0.79	0.87
Hydro	0.43^{*}	0.38	0.17^{***}	0.15^{***}
Pump	-0.01	-0.01	0.12^{*}	0.12
Wind	0.04^{**}	0.04^{*}	0.03^{***}	0.03^{***}

Table 9: ID15 IDA Spec C DiD with hourly Spain-France net cross-border flow added as a control. Same sample and specification as the ID15 IDA columns of Table 4; the right column of each pair adds the hourly cross-border net flow to the regression.

A.5 Per-CCGT-firm decomposition, both reforms

The pooled CCGT bid-shape DiDs of §6.1 mask substantial firm heterogeneity (Table 10), and the decomposition runs differently at the own-market DA15 day-ahead cell and the own-market ID15 intraday cell. On the DA15 day-ahead side, Naturgy is the dominant Cournot-style scale-up: α rises by EUR +70/MWh, β steepens by -0.40, and γ turns convex ($+0.0022$). Iberdrola is essentially flat on the level ($\alpha \approx 0$). EDP-HC has a small same-sign signature ($\alpha = +5$). On the ID15 intraday-auction side, Naturgy's β steepens and Iberdrola's α shows a sizeable but noisy level rise; the cross-firm pattern is consistent with intraday rebalancing being a broader, less Naturgy-concentrated phenomenon than day-ahead bidding. The pooled DiDs are unit-FE within-firm-pooled estimators that do not equal weighted firm-averages; what is interpretable here is the sign and the firm-ranking, not the level relative to the headline.

Firm	α	β	γ	Units
DA15 DA (own-market)				
Naturgy	69.9^{***}	-0.402^{***}	0.0022^{**}	17
Iberdrola	0.0	0.001	0.0000	10
Endesa	n.i.	n.i.	—	6
EDP-HC	5.2^{*}	-0.021^{*}	-0.0002	7
ID15 IDA (own-market)				
Naturgy	10.3	-1.197^{*}	0.0109	17
Iberdrola	32.2	-0.253	—	9
Endesa	-2.9	0.109	0.0044^{*}	7
EDP-HC	1.7	-0.211^{**}	-0.0029	6

Table 10: Per-firm CCGT bid-shape DiD, own-market cells of both reforms. One regression per firm on the firm's own CCGT fleet (unit FE within-firm). The firm-by-firm estimator does not net out firm-level intercepts, so magnitudes are not directly comparable to the pooled headlines; sign and ranking are interpretable. "n.i." marks the Endesa DA15 DA cell where the small unit count (6) and uneven within-fleet activation produce numerically unstable coefficients of order EUR +600/MWh on α; we suppress them rather than report.

A.6 Bandwidth δ (supportive on the slope β and the curvature γ)

We anchor the bandwidth to the contested margin of the market: $\delta = p_{90} - p_{50}$ of the MCP distribution per (window, market) (§4.3). The p_{90} choice separates the competing tier from the parked tier above the clearing-price ceiling (Appendix C); including the parked tier would mix non-strategic prices into σ_p, inflating it through a cross-tier spread. Table 11 reports the slope β and curvature γ DiD at three progressively wider bandwidths ($\delta = 100, 140, 200$ EUR/MWh) for the two headline own-market cells; the window-specific p_{90} headline row is included for comparison.

	ID15 IDA			**DA15 DA**		
δ	CCGT	Hydro	Pump	CCGT	Hydro	Pump
Slope β (EUR/MWh per MW)						
Headline (p_{90})	-0.881^{*}	-1.498	0.071	-0.150^{***}	-1.785^{***}	0.008
$\delta = 100$	-0.254^{***}	-0.781	0.035	-0.388^{***}	-0.636^{***}	0.012
$\delta = 140$	-0.234^{***}	-0.613	0.010	-0.493^{***}	-0.015	0.007
$\delta = 200$	-0.245^{***}	-0.862	-0.007	-0.481^{***}	-0.024	0.001
Curvature γ (EUR/MWh per MW^2)						
Headline (p_{90})	0.0100	-0.1088	0.0014	0.0010^{***}	-7.4938^{***}	0.0001
$\delta = 100$	0.0309^{**}	0.1572	0.0008	0.0057^{***}	-0.0591	0.0001^{***}
$\delta = 140$	0.0761	0.1459	0.0006	0.0059^{***}	0.0009	0.0002^{**}
$\delta = 200$	0.0761	0.1541	0.0007	0.0050^{***}	-0.0004	0.0001^{**}

Table 11: Bandwidth robustness for Spec C bid-shape DiD. Per-curve DiD coefficient θ on the slope β and curvature γ at progressively wider bandwidths. Window-specific p_{90} headline values: $\delta = 62$ for ID15 IDA, $\delta = 50$ for DA15 DA. CCGT β and γ are the most robust: DA15 DA CCGT β steepens monotonically with wider bandwidths ($-0.15 \to -0.49$) and γ stays convex and significant ($+0.001 \to +0.006$). ID15 IDA CCGT β steepens uniformly; γ is positive throughout but only sometimes significant. Hydro effects are narrow-band artefacts: DA15 DA hydro γ collapses from -7.6 at p_{90} to near zero at $\delta = 200$. α rows omitted (qualitatively similar to β and γ on the CCGT cells).

The IDA bandwidth pools the three sessions; session 3 has a materially wider $p_{90} - p_{50}$ than sessions 1–2, but a session-specific δ cannot flip signs — the per-session BSTS in §A.7 shows the price-level effect is uniform across sessions.

A.7 Per-IDA-session BSTS and the solar-coefficient symmetry test

The long pre-window of §5.1 earns its specification two ways: real and placebo absorb the spring solar surge through the same covariate channel (Table 12), and the per-session decomposition reproduces the pooled estimate (Table 13). The two tables together pin down the ID15 price effect at ∼EUR −40 to −50/MWh and show why a winter-only pre-window would have inflated the estimate to twice that.

Why the symmetry test matters. The placebo-net reading $\bar{\tau}_{\text{real}} - \bar{\tau}_{\text{placebo}}$ identifies a reform effect only if the BSTS treats the same calendar surge consistently across the two posteriors. The covariate channel for solar is the one most exposed to mis-identification, because solar output rises sharply between the late-winter pre-window and the spring post-window in every recent year. If the pre-window covers a full year of solar variation, the BSTS solar coefficient is identified across the full seasonal range and the post-window surge is absorbed through the covariate; the residual is the reform effect. If the pre-window is winter-only, the coefficient is identified off a narrow range and the surge partly bypasses the covariate, loading on the local-level component instead.

Pre-window	**Arm**	n_{pre}	**Effect (EUR/MWh)**	**Solar** $\hat{\beta}$	**Incl. prob.**	**Solar** Δ **(GWh/d)**
Long (Jun 2024 onwards)	Real 2025	278	**−38.3*****	−0.354	**1.00**	+15.4
	Placebo 2024	279	−18.1*	−0.273	**1.00**	+17.1
	Placebo 2026	278	−2.4	−0.400	**1.00**	+35.3
Short (Dec 2024 onwards)	Real 2025	80	**−78.8*****	−0.213	**0.48**	+48.7
	Placebo 2026	80	−8.9	−0.298	0.73	+80.9

Table 12: Solar-coefficient symmetry test for the ID15 IDA-price BSTS. Inclusion probability is the share of MCMC draws in which the spike-and-slab prior retains the solar regressor; solar Δ is the mean solar generation in the post-window minus the pre-window mean.

Per-session decomposition. With the long pre-window in place, the per-IDA-session BSTS reproduces the pooled estimate within a few EUR/MWh and confirms session symmetry (Table 13). All three sessions land at ∼EUR −45/MWh; the long-window 2026 placebo runs flat at ∼EUR −10/MWh per session, so the per-session placebo-nets cluster around −37 EUR/MWh and remain marginally significant on the joint test. The earlier ∼EUR −80/MWh per-session estimates reported in our draft of this table used the short post-ISP15 pre-window inherited from the 2026-placebo session-comparability constraint; in light of Table 12 we read them as inflated by the same channel and report the long-window numbers instead. The 2024 placebo for ID15 is not session-comparable (the pre-IDA-reform regime had six sessions rather than three), so we report it only at the pooled level.

Reform	Session	Effect (real)	Placebo 2024	Placebo 2026	Placebo-net (vs 2024)
ID15 IDA (long pre)	S1	**−48.4*****	—	−11.0	—
	S2	**−41.6*****	—	−8.4	—
	S3	**−47.6*****	—	−8.4	—
	Pooled	**−38.3*****	−18.1*	−2.4	−20.3
DA15 IDA	S1	15.4***	3.9	—	11.5**
	S2	14.4***	−1.8	—	16.2***
	S3	19.0***	10.6**	—	8.4

Table 13: Per-IDA-session and pooled BSTS on daily clearing prices. ID15 uses the long post-IDA-reform pre-window (2024-06-14 onwards); the 2024 placebo predates the 6 → 3 session reform so per-session ID15 rows show only the 2026 placebo. DA15 uses the reforzada-constant pre-window (2025-04-28 onwards) and a 2024 same-calendar placebo.

A.8 CCGT offer-type composition check (supportive)

The three OMIE structural offer types introduced in §4.2 (*simple*, *minimum-income condition* (MIC), and *block*) raise a potential mechanical confound for the DA15 CCGT σ_p widening of §6.1: a block offer is by construction a single tranche per period, so $\sigma_p = 0$ and $N_{\text{eff}} = 1$ mechanically. If CCGT's mix of structural types shifted across the reform, part of the pooled σ_p DiD could be a composition effect rather than within-quarter strategic engagement.

Figure 9 addresses this directly. The CCGT in-band offer mix does shift across the reform, but in the direction that should *dampen* the pooled σ_p widening: the simple-offer share *drops* from $\sim$ 29% to $\sim$ 22%, the block share *grows* from $\sim$ 30% to $\sim$ 35%, and the MIC share is roughly stable. More block offers means more mechanical zeros in the pooled σ_p average. Despite this dampening, the pooled σ_p DiD is still 1.25** EUR/MWh; restricting to simple offers only, the effect rises to 3.53*** (right panel). The σ_p widening of §6.1 is therefore real within-tranche behavior on the simple-offer arm, not a composition artefact — and the pooled headline is a conservative read.

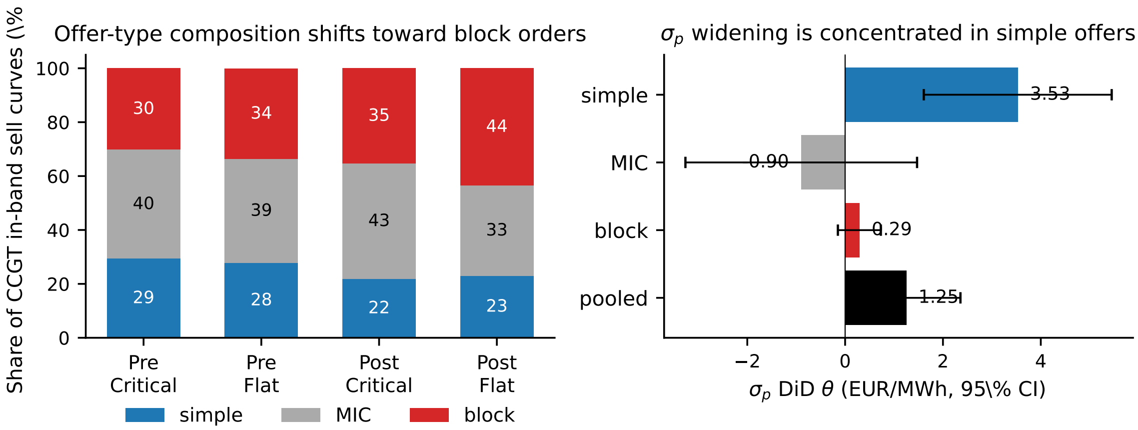

Figure 9: CCGT offer-type composition shift and σ_p DiD by offer type. *Left*: composition of CCGT in-band sell curves, pre vs post DA15, critical vs flat. *Right*: Spec C σ_p DiD by offer type with 95% CI.

A.9 Nuclear PDBF shortfall before the blackout

The §6.5 Spec A table uses auction-cleared MW (PDBC for DA, PIBCI for IDA) by construction, because OMIE auction programmes are the relevant object for the granularity reforms. For the blackout-context discussion in §7 we re-run the same BSTS on *auctions plus bilateral* dispatch — PDBF on the DA side (PDBC plus bilateral executions, the relevant baseline before any REE intervention) and PHF on the IDA side (the final post-IDA total dispatch) — because nuclear's bilateral leg carries a substantial share of its volume and is missed by the auction-only outcomes. Table 14 reports the result.

Reform window	Outcome	Real	Placebo 2024	Placebo-net
ID15 (Mar 19 – Apr 27, 2025, pre-blackout)	PDBF (DA + bilateral)	**−185.0*****	−41.8	**−143.2*****
	PHF (post-IDA total)	**−155.0*****	+14.7	**−169.6*****
DA15 (Oct 1 – Nov 9, 2025, reforzada-constant)	PDBF	−77.1**	−64.5***	−12.6
	PHF	−67.9**	−61.2***	−6.7

Table 14: Nuclear PDBF and PHF BSTS, ID15 and DA15. GWh/day. Same construction as Table 5; placebos are the same-calendar 2024 windows.

Reading. The ID15 placebo-net of −143 GWh/day in PDBF (and −170 in PHF) is the excess shortfall beyond the same-calendar 2024 spring refuelling pattern. The DA15 placebo-net brackets zero. The asymmetric placebo behaviour — large at the ID15 window, null at the DA15 window — supports a market-driven reading: in the 40-day post-ID15 pre-blackout window, daily-mean Spanish DA prices were near EUR 20/MWh with several days at or below zero (§6.5), and inflexible nuclear baseload had limited room to operate above the routine refuelling baseline. By autumn 2025 the price compression had reversed and the post-blackout regime kept nuclear synchronised for voltage support, so the same-calendar 2024 placebo and the real DA15 estimate move together and the net cancels.

A.10 BSTS on the bid-curve intercept α as a level outcome

Spec C identifies within-day redistribution (critical hours moving relative to flat hours), but the intercept α is the one bid-shape coordinate that is also a meaningful *level* object — the price at the top of the in-band ladder. To check the Spec C α reading with a complementary identification, we aggregate α to a daily mean per (technology, market) on a long pre-window (2022-01-01 onwards) and run a BSTS counterfactual with year-by-year wind and solar interactions and the same DA15 same-calendar 2024 placebo as the price BSTS (Table 15). The DA15 post-window is around four months long, which stretches the local-level state several quarters out from the most recent pre-window observation; BSTS predictive bands widen monotonically with the post horizon, so the DA15 BSTS-α point estimates are best read as level-direction indicators rather than precision estimates.

Market	Tech	Arm	Real (EUR/MWh)	Placebo	Placebo-net
	CCGT	long-pre	−63.9	−27.0	−36.9
ID15 IDA	Hydro	long-pre	−78.5***	−67.7**	−10.8
	Pump-storage	long-pre	−102.4***	−70.3***	−32.1
	CCGT	long-pre	3.3	−0.7	4.0
DA15 DA	Hydro	long-pre	−3.8	14.3	−18.1
	Pump-storage	long-pre	47.6	68.0	−20.4

Table 15: BSTS on daily-mean bid-curve intercept α, long pre-window with year-by-year renewable interactions. "Real" compares the post-reform window to the model's counterfactual; "Placebo" runs the same model on a same-calendar 2024 pre-reform window. The daily-mean α series is more variable than the auction-clearing price, so CIs are wide (omitted) and BSTS power is materially below the within-day Spec C DiD.

Reading. BSTS-on-α adds two pieces of evidence and one explicit null. (i) ID15 IDA hydro and pump-storage show a roughly EUR −10 to −30/MWh placebo-net level drop in α — a small, same-direction independent corroboration of the Spec C ID15 IDA hydro/pump α readings. (ii) ID15 IDA CCGT is null and noisy. (iii) DA15 DA cells are all uninformative under BSTS: daily-mean CCGT α in particular swings by hundreds of EUR/MWh day-to-day, so the BSTS CIs span thousands of EUR/MWh. The Spec C DiD is much more powerful here because unit FE plus the critical-flat partition absorb the day-to-day daily-mean noise. BSTS-on-α is therefore a corroborating test for the ID15 IDA hydro/pump α reading rather than an alternative headline.

Restricted-pre DA15 cross-check. Restricting the DA15 pre-window to the reforzada-constant interval 2025-04-28 – 2025-09-30 (rather than 2022-01-01 onwards) tightens the BSTS substantially (Table 16). On the restricted-pre, DA15 DA CCGT α rises by EUR +19.9 ($p = 0.031$) post-reform — in level. But the same-calendar 2024 placebo runs EUR +61.0 ($p < 0.001$): the autumn 2024 α rise was larger than the autumn 2025 rise, despite no reform between 2024-Apr-28 and 2024-Sep-30. The placebo-net is negative (−41.1). *The reform did not raise the daily-mean α of CCGT in level* — the Spec C α DiD of EUR +30.3/MWh is a within-day *critical-relative-to-flat* widening, not a level shift, which is exactly the object that Spec C is designed to identify and exactly the object that BSTS is the wrong tool for.

Tech	**Real 2025** (EUR/MWh)	**Placebo 2024**	**Placebo-net**
CCGT	19.9^{**}	61.0^{***}	−41.1
Hydro	−9.0	−23.3	+14.3
Pump-storage	25.6	45.3^{***}	−19.7

Table 16: BSTS on daily-mean α, DA15 with restricted reforzada-constant pre-window (2025-04-28 – 2025-09-30). 2024 same-calendar placebo. The negative placebo-nets confirm that the daily-mean α did not move beyond the same-calendar 2024 baseline at DA15 — the within-day Spec C α DiD identifies a critical-vs-flat redistribution, not a level rise.

A.11 Demand-side bid shape: a symmetric Spec C check

The headline Spec C applies to sell offers; the OMIE feed also exposes the buy side, with each buy unit carrying a structural agent type (retailer, direct industrial consumer, customer-of-last-resort distributor, portfolio aggregator, pump-storage charging-mode buyer). Running the same per-curve OLS (with tranches sorted descending in price so that β has the opposite sign convention than on the sell side) and the same critical/flat DiD on the four headline (reform, market) cells produces Table 17. Pump-buy unit FE work on a small but informative set; the customer-of-last-resort and distributor classes have too few units after FE to identify in all cells (rows suppressed).

Demand class	ID15 IDA	ID15 DA	DA15 DA	DA15 IDA
Intercept α (EUR/MWh)				
Retailer	−10.6**	5.4	15.0***	33.1***
DirectCons	0.2	—	−3.1	19.5***
PumpBuy	0.3	−4.9	−2.1*	0.8
Slope β (EUR/MWh per buy MW; on a descending price-ordered curve)				
Retailer	−1.688	−1.624	2.607***	−2.272**
DirectCons	−5.308***	—	−3.746***	1.555***
PumpBuy	−0.012	0.133	0.039***	0.017***
Curvature γ (EUR/MWh per buy MW^2)				
Retailer	−6.1746	−0.7050	2.7389	4.9286
DirectCons	−25.4***	—	−5.7976*	−0.0206
PumpBuy	−0.0002	0.0002	0.0008**	0.0005

Table 17: Demand-side Spec C DiD, critical vs. flat, unit FE. Per-curve OLS on the buy side with tranches sorted descending in price; β measures how price falls as cumulative buy MW expands down the demand ladder. Retailer = competitive supplier (BIG-4 dominated); DirectCons = large direct industrial buyer; PumpBuy = pump-storage in charging mode.

Reading. The demand-side α moves *up* at DA15 on both the day-ahead and the intraday-auction leg — retailers willing to pay EUR +14/MWh more at the top of the day-ahead demand ladder and EUR +33/MWh more at the intraday-auction ladder in critical relative to flat hours post-reform; direct consumers move by EUR +20/MWh at the IDA. The DA15 demand-side α rise sits next to the headline sell-side α rise of EUR +28/MWh for CCGT in the same DA cell: the cleared price in the contested band must reconcile sellers asking more with buyers willing to pay more — consistent with a Cournot-style equilibrium where both sides re-coordinate at a higher reservation price, not a one-sided supply shift. Pump-storage charging-mode buyers respond to DA15 with a flatter buy curve ($\beta > 0$, $\gamma > 0$ convex) consistent with the same operational logic as pump-storage on the sell side: greater willingness to absorb energy through the cheap end of the demand ladder once 15-min granularity allows finer-grain arbitrage of intra-hour price variation. ID15 IDA retailers run in the opposite direction (EUR −11/MWh on α), corroborating the ID15 IDA equilibrium-price compression documented in §6.5.

A.12 Demand-side buy share: who is two-sided in the IDA

The sell-side analysis of §6.2 reports the in-band sell-share for each production technology — a measure of how much of each tech's in-band activity is sell rather than buy. The natural mirror is the demand-side in-band buy-share: of each demand-side agent class's in-band MW, how much is buy versus sell. Table 18 reports the same three pre/post windows used in the bid-shape analysis. Buy-share is computed as buy MW/(buy MW + sell MW) pooled over the window, with the in-band region defined by the uniform bandwidth $|p_k - \text{MCP}| \leq 60$ EUR/MWh.

Market	Demand class	ID15 pre	DA15 pre	DA15 post
DA	Retailer	0.946	0.990	0.986
	DirectCons	1.000	1.000	1.000
	CUR	1.000	1.000	1.000
	Distrib	1.000	1.000	1.000
	PumpBuy	1.000	1.000	1.000
IDA	Retailer	0.646	0.618	0.627
	DirectCons	0.748	0.845	0.891
	CUR	0.390	0.466	0.339
	Distrib	0.406	0.000	0.000
	PumpBuy	0.767	0.870	0.844

Table 18: Demand-side in-band buy-share by class and market. Each cell is the pooled within-window buy MW / (buy MW + sell MW), computed on in-band tranches with $|p_k - \text{MCP}| \le 60$. Higher = more one-sided buying. ID15 pre = 2024-06-14 to 2025-03-18; DA15 pre = 2025-04-28 to 2025-09-30; DA15 post = 2025-10-01 to 2026-03-06.

Reading. The DA market is almost completely one-sided on the demand side: every class but Retailer is a pure buyer with buy-share = 1 across all three regimes, and Retailers (who can also resell) sit between 0.95 and 0.99. The IDA picture is qualitatively different: every class is materially two-sided. Distributors flip from ∼0.41 buy in the ID15 pre to fully selling (buy-share = 0) once we cross into the DA15 windows — they have become net sellers in the IDA contested band, presumably balancing a daytime overbuy from the day-ahead. Retailers stay stably two-sided around 0.62–0.65. Pump-storage charging buyers rise from 0.77 to 0.87 buy-share post-ID15, then drift back to 0.84 post-DA15 — using IDA more aggressively for charging after the intraday reform. CUR units swing the most dispersedly: 0.39 → 0.47 → 0.34. The takeaway is that the IDA two-sided participation pattern on the demand side is real, sizable, and shifts with each reform — it is not just production units that arbitrage across IDA-buy and IDA-sell.

B Continuous-market response at the three structural reforms

Specification. BSTS counterfactual on daily ES-leg continuous-market series with wind+solar+gas covariates and a weekly seasonal cycle (same setup as §5.1). Quantity outcomes are reported in *energy* (MWh per trade × delivery hours, summed per day) rather than raw MW sums — post-MTU15 each clock-hour holds four 15-min contracts where pre-MTU15 it held one 60-min contract, so a raw MW sum would be mechanically inflated. The volume-weighted price is energy-weighted. Three reform cutovers (ISP15 2024-12-11, ID15 2025-03-19, DA15 2025-10-01), each with a 2023/2024 same-calendar placebo.

Reform	Outcome	Real	Placebo	Placebo-net
ISP15	n_{trades}/day	$-3{,}120^{***}$	$-1{,}231^{***}$	$-1{,}889$
	GWh/day	-3.85^{*}	-2.23^{**}	-1.62
	VW-price (EUR/MWh)	-15.1	-27.4^{***}	12.3
ID15	n_{trades}/day	$\mathbf{22{,}335}^{***}$	$1{,}381^{*}$	$\mathbf{20{,}954}^{***}$
	GWh/day	-7.19^{***}	6.72^{***}	$\mathbf{-13.91}^{***}$
	VW-price (EUR/MWh)	-80.1^{***}	-14.1^{***}	$\mathbf{-66.0}^{***}$
DA15	n_{trades}/day	$5{,}018^{**}$	257	4,761
	GWh/day	3.85^{***}	-1.35	5.20^{**}
	VW-price (EUR/MWh)	4.7	-4.7	9.4

Table 19: Continuous-market response at the three structural reforms. BSTS posterior means with joint independent-posterior stars on Placebo-net.

B.1 ISP15 (settlement-only): a methodological null

ISP15 changed how REE settles imbalance positions (60→15 min) but left the OMIE continuous-market contract grid unchanged — traders still bought and sold hourly delivery contracts. A null effect is the right prediction here. Raw trade count and energy turnover both fall, but the same drift appears in the 2023 placebo (settlement bookkeeping innovations are not the source), and all three placebo-net intervals bracket zero. A settlement-only reform shows nothing on contract-structure outcomes — the specification reads the reform where it lives.

B.2 ID15 (60→15 min): the structural continuous-market reform

Pre-reform, each clock-hour of delivery hosted a single 60-min contract; post-reform, four 15-min contracts. The three outcomes move accordingly:

- *Trade count.* Each hour now offers four matching opportunities instead of one, so the trade-count surge is mechanical at first order — but informative: it confirms participants actually *used* the new finer grid rather than continuing to aggregate to the hourly contract.
- *Energy turnover.* Placebo-net is *negative*: more contracts, each smaller, and total energy changing hands drops. Two reasons: some hourly counterparties drop out when they don't see a 60-min match on offer, and some dispatchable bidding migrated into the now-finer IDA auctions instead (§6.4).
- *Volume-weighted price.* The continuous market clears in equilibrium with the auctions, so the ID15 cutover price drop on auction-cleared prices (§6.5) shows up here too — the cross-market spillover documented for ID15.

B.3 DA15: a smaller, indirect continuous-market effect

DA15 does not touch the continuous-market contract grid (already 15-min post-ID15); only the day-ahead *reference price* that continuous traders watch becomes finer. With a finer DA reference, traders reposition more aggressively in continuous: daily GWh up (joint 95% retains), trade count borderline, price effect statistical noise. DA15's continuous-market footprint is incremental and proportional to how much tighter the day-ahead reference now is — not a structural break, as expected from a reform that touches the reference market but not the continuous-market contract architecture.

C Market-structure context

C.1 The two-tier CCGT bid curve and the day-ahead/intraday contrast

Every Big-4 CCGT fleet runs a two-tier day-ahead offer: a low *competing* tier near marginal cost and a high *parked* tier above the clearing-price ceiling. Withholding is firm-wide, not a Naturgy story. The same fleets bid a single competitive tier with negligible withholding in the intraday auction, after Fase I has cleared. A unit's marginal cost does not change between two auctions hours apart. The parked tier is a placeholder rather than a Fase I bid: Fase I up-redispatch is settled through a separate restriction offer submitted to REE.[8] MTU15-DA grants CCGT a within-hour degree of freedom over its competing tier; the CCGT auction-cleared scale-up in §6.5 is the competing tier expanding into the four quarters of each peak hour.

C.2 Market concentration migrated from wholesale to ancillary services

Wholesale-market HHI hovers around 1 100 over 2016–2024 (below the moderate-concentration threshold of 1 500) and falls mid-day when solar peaks (solar is less concentrated than the Big-3). Day-ahead restrictions upward HHI above 4 000; deviations upward above 6 000; tertiary upward around 5 000. Median of 4 firms with accepted upward restrictions bids. As renewables eroded wholesale-market power, technical and locational requirements concentrated the ancillary markets that absorb their variability.

C.3 The reforzada cost increase concentrates on CCGT

The post-blackout REE intervention is technology-specific. Decomposing weekly dispatch into its components — the day-ahead-cleared base (PDBC), bilateral executions, and the post-DA top-up routed through IDA and REE real-time redispatch — the CCGT panel of Figure 10 shows the day-ahead-cleared layer (blue) collapsing post-blackout while the IDA+REE-RT layer (red) balloons. Other technologies are largely flat across the same window.

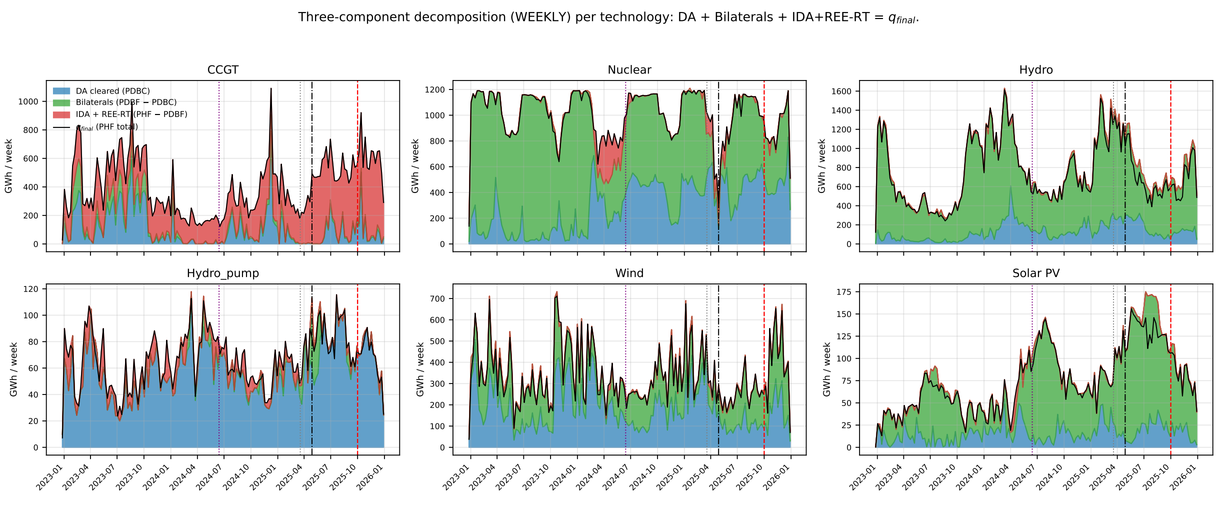

Figure 10: Weekly three-component decomposition of dispatch, by technology. PDBC (blue, day-ahead-cleared) + bilateral executions (green) + IDA + REE real-time top-up (red) = final post-IDA dispatch (black). Vertical guides: MTU15-IDA, April 2025 blackout, MTU15-DA.

[8] Procedure PO 3.2, settlement PO 14.4 ap. 20.1; see also CNMC (2025c,a). Earlier drafts conflated the two; the distinction justifies measuring bid shape and bid levels in the competing tier only, which is what the window-specific p_{90} bandwidth in §4.3 achieves.

This per-technology pattern is the cleanest evidence that the reforzada cost increase is a separate policy effect from the granularity reforms: a granularity reform that touched all bid-side firms equally would not produce a CCGT-only intervention pattern. The day-ahead CCGT cleared scale-up of §6.5 (DA15 only, critical hours) sits alongside this REE-intervention story: both are CCGT-side responses to the operational environment, but the granularity-driven response is competitive-clearing (and consumer-favourable) while the reforzada-driven response is REE-paid redispatch.

C.4 Representative bid curves by technology

Figures 11–12 show real day-ahead bid step functions for one representative unit per technology, at an evening-ramp critical-hour quarter (Figure 11) and at a flat overnight hour on a windy night (Figure 12). The visual contrast supports the §5.3 exclusion of solar from Spec C and the §6.7 reading of wind's small bid-shape DiD coefficients: dispatchables structure their competing tier inside the in-band region; renewables submit single or near-single blocks far below the band, so a within-band σ_p or N_{eff} outcome is mechanically zero or one. Wind aggregators do occasionally submit multi-tranche micro-ladders below the band — the WMVD088 panel in Figure 12 shows 15 tranches over the narrow $[-20, +1.5]$ EUR/MWh range, consistent with the operational (imbalance-bill) rather than strategic incentive that drives wind's bidding (§6.7) — but these micro-ladders sit outside the in-band region and so do not enter the Spec C outcomes.

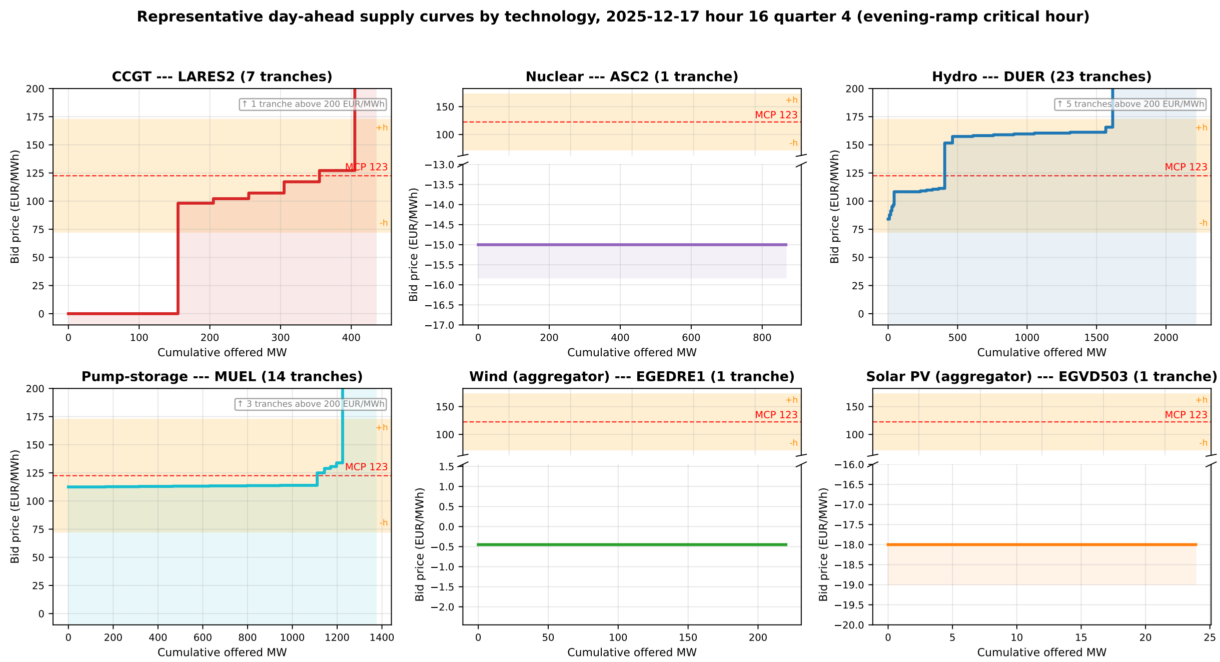

Figure 11: Representative day-ahead supply curves by technology, 2025-12-17 hour 17 quarter 4 (post-MTU15-DA, evening-ramp critical hour). Red dashed: MCP; orange band: in-band region [MCP $\pm\,\delta$] with $\delta = 50$ (§4.3). Panels with bid ranges far below the band use a broken y-axis for readability.

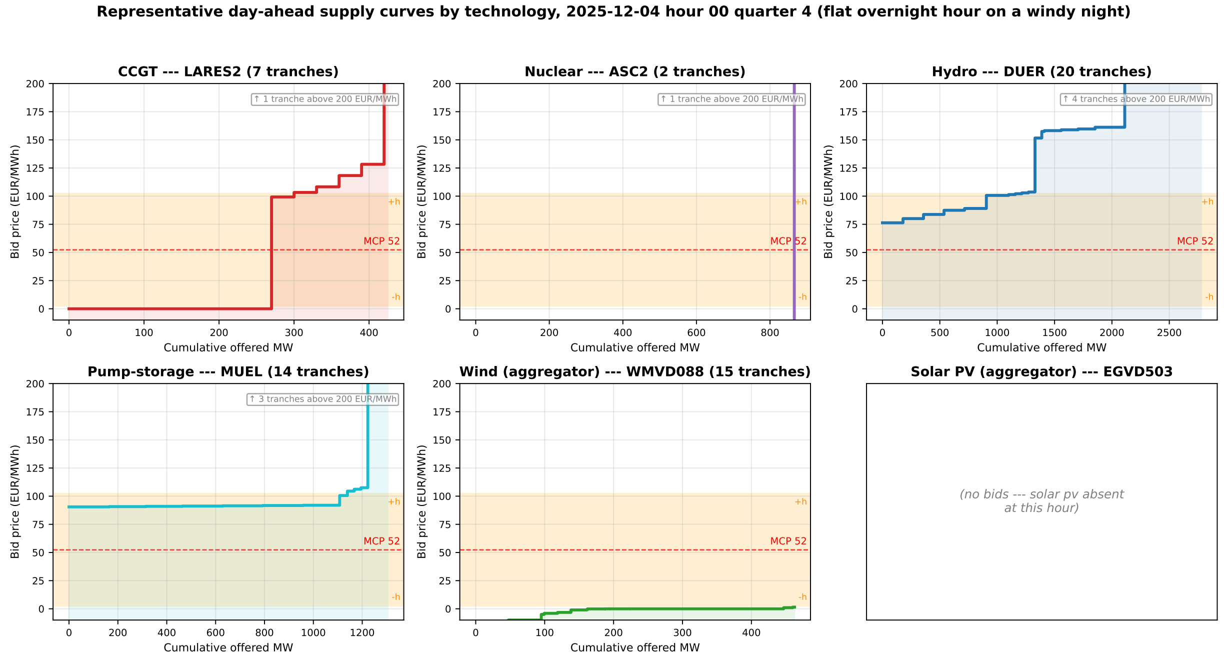

Figure 12: Representative day-ahead supply curves by technology, 2025-12-04 hour 01 quarter 4 (flat overnight hour on a windy night). Same colour scheme and broken y-axis convention as Figure 11. The wind panel (WMVD088) shows a 15-tranche aggregator micro-ladder concentrated in $[-20, +1.5]$ EUR/MWh, well below the band.

Buy vs sell IDA bid shapes (pre-ID15). Figure 13 shows representative IDA bid curves with the sell side (blue, ascending) and the buy side (red, descending) overlaid for one unit per tech in the pre-ID15 window, when buy-side activity was largest (§6.3). The shapes differ sharply across techs: CCGT and pump-storage submit dense sell ladders with a thin one-tranche buy book (sellers with a small rebuy option); hydro and renewable aggregators write multi-tranche curves on *both* sides (forecast-driven rebalancing that can clear in either direction); nuclear barely participates on the buy side (the entire 9-month pre-ID15 window contains a single nuclear (date, session, period) cell with both sides). The renewable-aggregator panels are the clearest illustration of the buy-side rebalancing role of IDA: the buy and sell curves intersect near the realised MCP, and the aggregator clears on whichever side its quarter-level forecast falls.

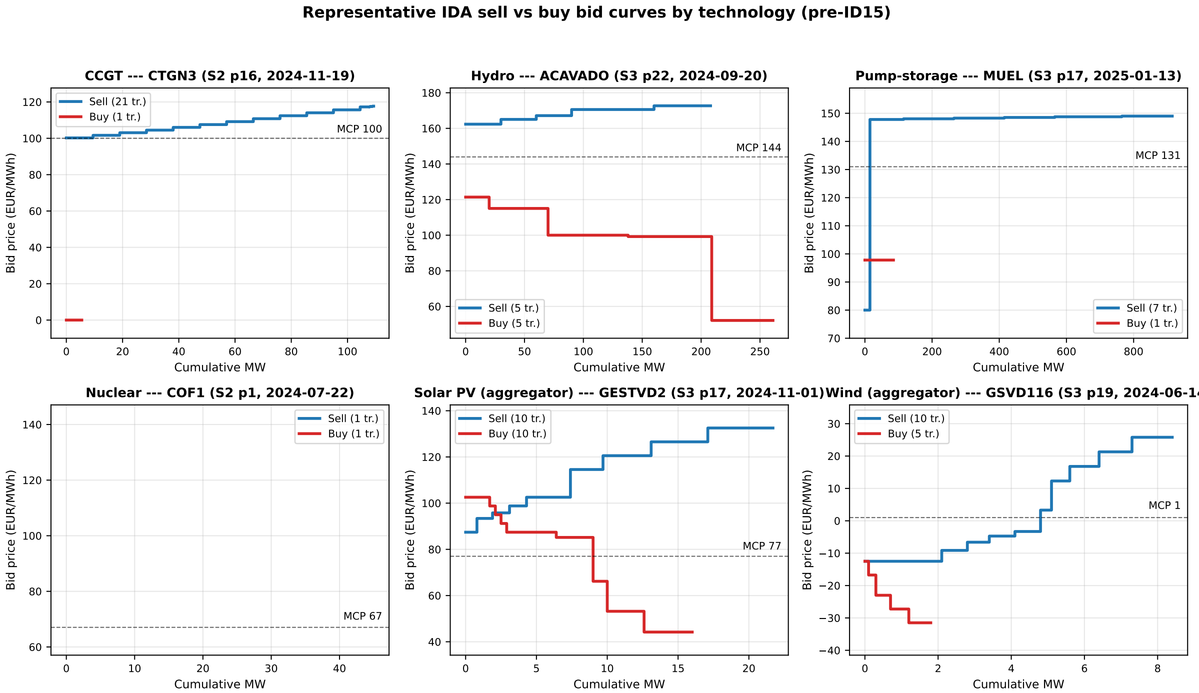

Figure 13: Representative IDA sell vs buy bid curves by technology, pre-ID15. One representative (unit, date, session, period) per tech. Blue: sell curve (supply, ascending in price). Red: buy curve (demand, descending in price). Dashed: MCP. Pre-ID15 window 2024-06-14 to 2025-03-18.

D Theoretical derivations

D.1 Setup details and information arrival

This subsection records the timing of information arrival, the Ito and Reguant residual-demand specification used throughout, and how that specification specialises when one or both markets are hourly.

Timing. Stage 1 (DA): only $\bar{A}$ and the distribution of μ_t are known to the strategic bidder; the aggregator knows $\bar{Y}_h$, $\{\bar{y}_t\}$ and the distribution of η_t. Stage 2 (ID): μ_t has realised and is common knowledge to first order. Stage 3 (balancing): η_t realises and settlement occurs at the balancing reference.

Residual demand (Ito and Reguant (2016) form). The strategic residual monopolist with constant marginal cost c serves what the operational aggregator does not cover, net of any arbitrage position s_t carried from forward to spot. Per delivery product, with all quantities in MWh:

$$q_{1t} = \bar{A}_t - b_{1t}\, p_{1t} - s_t - S^{op}_{1t}, \qquad q_{2t} = b_{2t}\,(p_{1t} - p_{2t}) + s_t - S^{op}_{2t}, \tag{4}$$

where $\bar{A}_t = \bar{A} + \mu_t$ is the per-subperiod demand intercept, S^{op}_{mt} is the aggregator's offered MWh in market $m \in \{1, 2\}$ at subperiod t (uniform S^{op}_m/Q when market m is hourly), and b_{mt} is market thinness (with per-subperiod asymmetry allowed). We use p_{mt} throughout for the per-subperiod price; when market m is hourly, $p_{mt} \equiv p_m$ is constant in t (the auction clears one price for the whole hour). The within-hour demand shock satisfies $\mathbb{E}[\mu_t] = 0$ and $\mathrm{Var}(\mu_t) = \sigma^2_\mu$; the aggregator's per-subperiod realisation is $y_t = \bar{y}_t + \eta_t$ with forecast-error variance σ^2_η.

Hourly residual demand by specialisation. When market m is hourly the bidder commits to the auction at one price for the hour, against the *ex-ante expected* hourly aggregate of residual demand. The general per-subperiod object (4) specialises by setting $\bar{A}_t \equiv \bar{A}$ (the auction-time information set: $\mathbb{E}[(1/Q)\sum_t \mu_t] = 0$), $b_{mt} \equiv b_m$, $p_{mt} \equiv p_m$, and $S^{op}_{mt} \equiv S^{op}_m/Q$. The realised within-hour variation in μ_t that this substitution suppresses does not vanish from the model: it loads onto the balancing residual (§D.7) and shows up in the system imbalance cost C^I_{sys} in the welfare derivations (§D.9). In $(1,1)$ both equations reduce to one product apiece; in $(1,Q)$ only the spot demand keeps the per-period index.

The two-stage analysis here covers the forward-spot auction pair. The balancing market enters only as the ex-post settlement reference; we collapse it into the balancing discrepancy Δ_t used in §D.7.

D.2 Operational aggregator: full scheduling problem

We solve the aggregator's full scheduling problem in four steps: the problem statement, the closed-form equilibrium when both markets are hourly, the equilibrium under spot-per-period scheduling, and the comparative static $\partial\Delta^*/\partial\sigma^2_\eta > 0$ that closes (2).

D.2.1 Problem statement

The aggregator chooses $(S^{op}_{1t}, S^{op}_{2t})_{t=1}^{Q}$ to solve

$$\max_{(S^{op}_{1t}, S^{op}_{2t})_t} \; \mathbb{E}\left[\sum_t \left(p_{1t}\, S^{op}_{1t} + p_{2t}\, S^{op}_{2t}\right) \;-\; \gamma\,\mathcal{I}\right] \quad \text{s.t.} \quad \sum_t (S^{op}_{1t} + S^{op}_{2t}) = \bar{Y}_h, \tag{5}$$

with quadratic imbalance penalty $\mathcal{I} = \sum_t (y_t - S^{op}_{1t} - S^{op}_{2t})^2$ under per-period settlement and uniform-across-t allocation in any market that is hourly.

D.2.2 Equilibrium under hourly schedules ($(1,1)$ and $(1,Q)$ with hourly aggregator)

With both markets hourly the aggregator chooses scalar (S^{op}_1, S^{op}_2) with $S^{op}_1 + S^{op}_2 = \bar{Y}_h$. The per-subperiod implementation is uniform: $S^{op}_{1t} = S^{op}_1/Q$ and $S^{op}_{2t} = S^{op}_2/Q$. Per-subperiod imbalance under per-period settlement is

$$\mathcal{I} \;=\; \sum_t \left(y_t - (S^{op}_1 + S^{op}_2)/Q\right)^2 \;=\; \sum_t \left((\bar{y}_t - \bar{Y}_h/Q) + \eta_t\right)^2,$$

which depends on $(\bar{y}_t - \bar{Y}_h/Q)$ and η_t, not on the forward-vs-spot split. The aggregator's allocation is therefore driven purely by the price differential: it commits at forward if $p_1 > p_2$ and is indifferent if $p_1 = p_2$. We normalise $S^{op}_1 = S^{op}_2 = \bar{Y}_h/2$ as a tie-break for the full-arbitrage benchmark.[9]

D.2.3 Equilibrium under DA-hourly + ID-granular (state $(1,Q)$)

Notation for forecast resolution. The within-hour heterogeneity that loads onto the aggregator's imbalance has two components: a structural per-subperiod expected production variation $(\bar{y}_t - \bar{Y}_h/Q)$, known already at Stage 1, and a forecast update η_t resolved between Stages 1 and 2 (i.e. between the forward close and the spot close). η_t has variance σ^2_η and is the empirically relevant "forecast-error shock": bigger σ^2_η means more within-hour production information that materialises after the forward allocation is fixed.

[9]The 50/50 split is innocuous for the comparative-statics path we trace. Under non-full-arbitrage regimes ((R1), (R3), (R4)) the equilibrium has $p_1 > p_2$ and the aggregator's optimum is the corner $S^{op}_1 = \bar{Y}_h$, but the (1,1) state's role in our argument is only as the baseline against which the (1,Q) and (Q,Q) reforms are compared, and the comparative statics in $\Delta^*(\sigma^2_\eta)$ and the wedge SD are not driven by which corner we pick at (1,1).

Optimal per-subperiod spot schedule. The aggregator chooses per-subperiod $\{S_{2t}^{op}\}_t$ at spot using its Stage-2 forecast $\bar{y}_t^{(2)} \equiv \bar{y}_t + \eta_t$ (the updated expected production at the time of the spot close). Setting

$$S_{2t}^{op*} \;=\; \bar{y}_t^{(2)} - \frac{S_1^{op}}{Q} \;=\; \bar{y}_t + \eta_t - \frac{S_1^{op}}{Q}, \tag{6}$$

matches the Stage-2 forecast net of the uniformly-allocated forward commitment. The realised per-subperiod imbalance under (6) reduces to the residual post-Stage-2 noise ε_t (independent of η_t, with small variance). The expressions below use $|\cdot|$ rather than the signed deviation: the imbalance penalty is paid on the *magnitude* of the schedule-vs-realisation gap, not on the signed quantity, so $\mathbb{E}|\varepsilon_t| > 0$ even though $\mathbb{E}[\varepsilon_t] = 0$. The expected aggregate imbalance under per-period scheduling is

$$\mathbb{E}[\mathcal{I}_{\text{granular}}] \;=\; Q \cdot \mathbb{E}[\,|\varepsilon_t|\,]\,.$$

Under the hourly schedule (§D.2.2) the aggregator must commit at Stage 1 using the Stage-1 forecast $\bar{y}_t^{(1)} = \bar{y}_t$; the Stage-1-to-Stage-2 update η_t then loads onto the imbalance:

$$\mathbb{E}[\mathcal{I}_{\text{hourly}}] \;=\; \sum_{t=1}^{Q} \mathbb{E}\big[\,\big|(\bar{y}_t - \bar{Y}_h/Q) + \eta_t + \varepsilon_t\big|\,\big]\,.$$

The aggregator's net gain from per-period spot scheduling is

$$G(\sigma_\eta^2) \;\equiv\; \gamma \sum_t \Big[\mathbb{E}\big|(\bar{y}_t - \bar{Y}_h/Q) + \eta_t + \varepsilon_t\big| \;-\; \mathbb{E}\big|\varepsilon_t\big|\Big], \tag{7}$$

strictly increasing in σ_η^2 (the avoided Stage-1-to-Stage-2 forecast shock) and in $\sigma_a^2 \equiv \mathrm{Var}(\bar{y}_t - \bar{Y}_h/Q)$ (the structural within-hour variation). Both components map empirically to within-hour heterogeneity: clock-hours where production swings within the hour carry larger σ_η^2 and larger σ_a^2.

D.2.4 Quantity shift $\Delta^*(\sigma_\eta^2)$ and its comparative static

Write $\Delta \equiv \bar{Y}_h - 2S_1^{op}$ for the aggregator's forward$\to$spot quantity shift (so $\Delta = 0$ is the 50/50 baseline and $\Delta > 0$ means more MWh allocated to spot). The aggregator's full Stage-1 problem is to maximise revenue net of the expected imbalance penalty,

$$\max_{S_1^{op}} \;\; p_1\, S_1^{op} + p_2(\bar{Y}_h - S_1^{op}) \;-\; \gamma \sum_t \mathbb{E}\big|(\bar{y}_t - (\bar{Y}_h - \Delta)/Q) + \eta_t\big|,$$

where the first two terms are revenue under the indifferent 50/50 baseline plus the marginal revenue effect of shifting $\Delta/2$ to spot, and the penalty term collapses to G from (7) when Δ is set to its optimum. Differentiating in S_1^{op} and using $\partial\Delta/\partial S_1^{op} = -2$:

$$(p_1 - p_2) \;=\; \gamma \cdot \frac{\partial}{\partial S_1^{op}}\Big[\sum_t \mathbb{E}\big|(\bar{y}_t - (\bar{Y}_h - \Delta^*)/Q) + \eta_t\big|\Big]\Big|_{\Delta^*}. \tag{8}$$

The LHS is the marginal revenue of one extra MWh held in forward; the RHS is the marginal imbalance cost of the same shift. Setting LHS = RHS pins down Δ^*. The right-hand side is monotone in Δ^* by convexity of the absolute-value expectation in its argument. Differentiating implicitly in σ_η^2,

$$\frac{\partial \Delta^*}{\partial \sigma_\eta^2} \;=\; \frac{-\,\partial^2 G/\partial S_1^{op}\partial\sigma_\eta^2}{\partial^2 G/\partial (S_1^{op})^2} \;>\; 0,$$

since the numerator is positive (a larger forecast-error variance increases the marginal cost of an extra forward-committed MWh, because the realised $|\eta_t|$ contribution to imbalance does not net out across subperiods under per-period settlement) and the denominator is positive by convexity. This establishes (2): the aggregator's quantity shift from forward to spot is strictly increasing in the forecast-error variance.

D.3 Strategic bidder: optimal two-tranche ladder

We derive the strategic bidder's closed-form two-tranche ladder under uniform-price clearing, the FOCs in (p^L, p^H, k^L), the equilibrium spread $\Delta p^* = \Delta_\mu/(2b)$, and a stylisation note that flags the simplification absorbed into the boundary condition.

D.3.1 Profit under uniform-price clearing

The strategic bidder is the residual monopolist; there is no higher-priced fringe bid above its top tranche. Under uniform-price clearing (as in OMIE day-ahead and intraday auctions), the clearing price is therefore set by the marginal accepted tranche of the strategic bidder's own ladder: p^L when only the low tranche clears, p^H when both clear. For a delivery product at which the residual-demand intercept is $\bar{A} + \mu$ (with $\mu \sim \text{Uniform}[-\Delta_\mu, +\Delta_\mu]$), the strategic bidder commits a ladder $\{(p^L, k^L), (p^H, k^H)\}$ with $p^L \le p^H$, $k^L + k^H = K$ *before* μ realises. We use b for the market thinness (b_{1t} or b_{2t}, depending on which market the ladder is on).

Crossover demand state μ^*. Residual demand at price p^L in demand state μ is $\bar{A} + \mu - b\,p^L$. The cutoff μ^* is the demand state at which this residual demand exactly equals the low-tranche volume:

$$\bar{A} + \mu^* - b\,p^L \;=\; k^L \qquad \Longleftrightarrow \qquad \mu^* \;\equiv\; k^L + b\,p^L - \bar{A}.$$

For $\mu \le \mu^*$ residual demand at p^L is at most k^L, so only the low tranche clears and the auction prices at p^L. For $\mu > \mu^*$ residual demand exceeds k^L at p^L, so both tranches clear and uniform pricing carries p^H on the full capacity K. The cleared MWh at μ is then

$$q(\mu) = \begin{cases} q^L \;=\; k^L & \text{if } \mu \le \mu^*, \\ K \;=\; k^L + k^H & \text{if } \mu > \mu^*. \end{cases}$$

The two-tranche reduction treats $q^L = k^L$ as constant on $\mu \in [-\Delta_\mu, \mu^*]$ rather than tracking the smooth residual-demand schedule below μ^*; this is the stylisation absorbed into the boundary condition (see closing note of this subsection). Expected profit per delivery product is

$$\mathbb{E}[\pi] \;=\; \frac{1}{2\Delta_\mu}\left[\int_{-\Delta_\mu}^{\mu^*} (p^L - c)\,q^L\,d\mu \;+\; \int_{\mu^*}^{\Delta_\mu} (p^H - c)\,K\,d\mu\right], \tag{9}$$

with $(p^H - c)\,K$ in the second integrand reflecting that all K units fetch p^H when the high tranche is marginal.

Implicit price cap via the demand support. Real auctions impose an explicit price cap $p^{\max}$ (around EUR 3000/MWh in MIBEL); the formulation above absorbs the cap into the demand support, since the bidder pushes p^H to the price that clears K at the highest demand realisation $\mu = \Delta_\mu$ (see FOC below). The two parameterisations are equivalent up to a constant rescaling and we keep Δ_μ for tractability.

D.3.2 First-order conditions

Take derivatives in (p^L, p^H, k^L) holding $K = k^L + k^H$ fixed. Define $w^L = k^L/K$, $w^H = k^H/K$.

FOC in p^H.

$$\frac{\partial \mathbb{E}[\pi]}{\partial p^H} \;=\; \frac{1}{2\Delta_\mu}\int_{\mu^*}^{\Delta_\mu} K\,d\mu \;=\; \frac{K(\Delta_\mu - \mu^*)}{2\Delta_\mu} \;>\; 0$$

unless $\mu^* = \Delta_\mu$. The derivative is strictly positive at any interior μ^*: raising p^H earns extra mark-up on the full capacity K in every demand state above μ^*, and (within the interior) carries no quantity loss because both tranches still clear. The bidder therefore pushes p^H to the boundary where even the highest demand state $\mu = \Delta_\mu$ just clears full capacity: residual demand at p^H equals K when $\bar{A} + \Delta_\mu - b\,p^H = K$, i.e.

$$p^H \;=\; (\bar{A} + \Delta_\mu - K)/b \quad \text{at the binding boundary.}$$

Substituting the symmetric-solution value $\mu^* = 0$ (derived below from the joint FOCs) gives the closed form $p^{H*} = (\bar{A} + bc)/(2b) + \Delta_\mu/(4b)$.

FOC in p^L.

$$\frac{\partial \mathbb{E}[\pi]}{\partial p^L} \;=\; \frac{1}{2\Delta_\mu}\left[\int_{-\Delta_\mu}^{\mu^*} q^L\,d\mu \;+\; (p^L - c)q^L \cdot \frac{\partial \mu^*}{\partial p^L}\right] \;=\; \frac{1}{2\Delta_\mu}\big[(\mu^* + \Delta_\mu)\,q^L \;+\; (p^L - c)q^L\,b\big] \;=\; 0.$$

Solving: $\mu^* + \Delta_\mu + b(p^L - c) = 0$, i.e. $\mu^* = -\Delta_\mu - b(p^L - c)$. Combined with the definition $\mu^* = k^L + bp^L - \bar{A}$, this gives $k^L = \bar{A} - bp^L + \mu^*$. At $\mu^* = 0$: $p^{L*} = (\bar{A} + bc)/(2b) - \Delta_\mu/(4b)$.

FOC in k^L. Differentiating (9) in k^L with $K = k^L + k^H$ held fixed ($\partial k^H/\partial k^L = -1$, $\partial K/\partial k^L = 0$), and using $\partial \mu^*/\partial k^L = 1$ from the definition of μ^*:

$$\frac{\partial \mathbb{E}[\pi]}{\partial k^L} \;=\; \underbrace{\frac{p^L - c}{2\Delta_\mu}\,(\mu^* + \Delta_\mu)}_{\text{direct: more MWh sold at } p^L \text{ in regime 1}} \;-\; \underbrace{\frac{p^H - c}{2\Delta_\mu}\,(\Delta_\mu - \mu^*)}_{\text{boundary: probability mass shifts from regime 2 to regime 1 as } \mu^*\uparrow} \;=\; 0.$$

The two terms have opposite signs: enlarging the low tranche earns extra inframarginal markup over $[-\Delta_\mu, \mu^*]$, but shifts probability mass out of the high-markup regime $(p^H - c) > (p^L - c)$. The optimum balances them. At the symmetric solution $\mu^* = 0$, the FOC becomes $(p^L - c) = (p^H - c)$ scaled by the $1/2$ probability of each regime — not the literal equation $p^L = p^H$, but rather p^L and p^H are chosen so that the marginal benefit of one extra low-tranche MWh exactly matches the marginal cost. With p^{L*} and p^{H*} pinned by their own FOCs, the joint solution requires $\mu^* = 0$ exactly: **$\mu^* = 0$ is the median demand cutoff** — the bidder structures the ladder so that exactly half the demand realisations clear only the low tranche and half clear both. Substituting $\mu^* = 0$ in $k^L = \bar{A} - bp^{L*} + \mu^*$ and the closed form for p^{L*},

$$k^{L*} \;=\; \bar{A} - b\left(\frac{\bar{A} + bc}{2b} - \frac{\Delta_\mu}{4b}\right) \;=\; \frac{\bar{A} - bc}{2} + \frac{\Delta_\mu}{4}.$$

Closed-form interior optimum.

$$\boxed{p^{L*} = \frac{\bar{A} + b\,c}{2b} - \frac{\Delta_\mu}{4b}, \quad p^{H*} = \frac{\bar{A} + b\,c}{2b} + \frac{\Delta_\mu}{4b}, \quad \mu^{**} = 0, \quad k^{L*} = \frac{\bar{A} - b\,c}{2} + \frac{\Delta_\mu}{4}.} \tag{10}$$

Comparative statics. From (10), the optimal pre-realisation tranche spread is

$$\Delta p^* \;=\; p^{H*} - p^{L*} \;=\; \frac{\Delta_\mu}{2b}. \tag{11}$$

Running the same argument on the hourly aggregate $\tilde{\mu} = Q^{-1}\sum_t \mu_t$ (variance $\sigma_\mu^2/Q < \sigma_\mu^2$) gives a strictly smaller optimal spread, $\Delta p^*|_{\text{hourly}} = \tilde{\Delta}_\mu/(2b)$ with $\tilde{\Delta}_\mu = \Delta_\mu/\sqrt{Q}$. The per-period ladder is therefore strictly richer than its hourly counterpart by a factor $\sqrt{Q}$.

Stylisation note. The closed-form (10) requires that the capacity K be tuned to the boundary condition p^{H*} binds at the highest demand realisation $\mu = \Delta_\mu$; for general K the FOC in p^L also picks up a term $-bK(p^H - c)$ from the second integrand of (9), and the optimum shifts off the symmetric form. The closed form is therefore a stylised approximation, with the auction microfoundation absorbed into the boundary condition. Comparative statics in Δ_μ and b are robust to this stylisation: Δp^* widens with the demand range and contracts with thinness.

Empirical measures of within-curve bid-shape dispersion (MW-weighted price SDs, effective tranche counts) are motivated by this widening but are not derived from the two-state reduction — they require richer bid-curve structure than the two-tranche approximation supports.

D.4 $(1,1)$ equilibrium

We solve the pre-reform game by backward induction: the Stage 2 best response (§D.4.1), then the four arbitrage regimes at Stage 1 (§D.4.2). The scarcity wedge $p_1 > p_2 > c$ survives in regimes (1), (3), and (4); only competitive full arbitrage closes it.

D.4.1 Stage 2 best response

The strategic bidder's Stage 2 problem, taking p_1, q_1 and s as given:

$$\max_{p_2} (p_2 - c) q_2 \quad \text{s.t.} \quad q_2 = b_2(p_1 - p_2) + s - S_2^{op}.$$

FOC: $b_2(p_1 - p_2) + s - S_2^{op} + (p_2 - c)(-b_2) = 0$, giving

$$p_2^*(p_1, s) \;=\; \frac{p_1 + c}{2} + \frac{s - S_2^{op}}{2\, b_2}. \tag{12}$$

D.4.2 Stage 1 best response and equilibrium in (p_1, s)

Substituting (12) into the strategic bidder's total profit and using $q_1 = \bar{A} - b_1 p_1 - s - S_1^{op}$,

$$\Pi(p_1, s) \;=\; (p_1 - c)(\bar{A} - b_1 p_1 - s - S_1^{op}) \;+\; (p_2^*(p_1, s) - c) q_2^*(p_1, s).$$

The four arbitrage cases pin s differently.

(R1) No arbitrage ($s = 0$). The strategic bidder solves a Cournot problem in each market separately. The forward FOC gives $p_1^* = (\bar{A} - S_1^{op} + b_1 c)/(2b_1)$. Combining with (12) yields $p_1^* > p_2^* > c$ and $q_1, q_2 > 0$ (Ito and Reguant Result 1).

(R2) Full arbitrage ($p_1 = p_2$). Setting $p_1 = p_2$ in (12) gives $s = b_2(p_1 - c) + S_2^{op}$. Substituting back, total cleared MWh $q_1 + q_2 = \bar{A} - b_1 p_1 - S_1^{op} - S_2^{op} = \bar{A} - b_1 p_1 - \bar{Y}_h$ (using $S_1^{op} + S_2^{op} = \bar{Y}_h$). The strategic bidder maximises $(p_1 - c)(\bar{A} - b_1 p_1 - \bar{Y}_h)$, giving

$$p_1^{*,F} \;=\; \frac{\bar{A} - \bar{Y}_h + b_1 c}{2b_1}, \qquad p_2^{*,F} \;=\; p_1^{*,F}. \tag{13}$$

Note that $s > 0$ *reduces* the strategic bidder's total output relative to no arbitrage (Ito and Reguant Result 2): even competitive arbitrage does not restore competitive prices, because the bidder withholds more in response.

(R3) Limited arbitrage ($s \le K$ binding). If K is below the full-arbitrage level, $s = K$ binds and $p_1 - p_2 > 0$. From (12), $p_1 - p_2 = (p_1 - c)/2 - (K - S_2^{op})/(2b_2)$; the wedge depends on K and on S_2^{op} (Ito and Reguant Result 3). The smaller K, the larger the residual wedge.

(R4) Strategic arbitrage. The arbitrageur maximises $\pi^A(s) = s(p_1 - p_2)$ subject to (12):

$$\pi^A(s) \;=\; s\left[\frac{p_1 - c}{2} - \frac{s - S_2^{op}}{2\,b_2}\right].$$

FOC in s:

$$\frac{p_1 - c}{2} - \frac{2s - S_2^{op}}{2\,b_2} \;=\; 0 \quad\Longleftrightarrow\quad s^{*,S}(p_1) \;=\; \frac{b_2(p_1 - c)}{2} + \frac{S_2^{op}}{2}.$$

Substituting into (12) (with $S_2^{op} \to 0$ for clarity in the wedge derivation),

$$p_2^{*,S} \;=\; \frac{p_1 + c}{2} + \frac{b_2(p_1 - c)/2}{2\,b_2} \;=\; \frac{p_1 + c}{2} + \frac{p_1 - c}{4} \;=\; \frac{3p_1 + c}{4}. \tag{14}$$

The forward-spot wedge is $p_1 - p_2^{*,S} = (p_1 - c)/4$, matching (1) in the main body (Ito and Reguant Result 4). The Stage 1 forward price $p_1^{*,S}$ is the maximiser of the strategic bidder's compound profit; algebra in the standard quadratic gives $p_1^{*,S} - c = 8(\bar{A} - b_1 c - \bar{Y}_h)/(16b_1 - b_2)$ at $S_2^{op} = 0$, $S_1^{op} = \bar{Y}_h$. The wedge $(p_1^{*,S} - c)/4 > 0$ persists at the equilibrium price.

D.5 $(1, Q)$ equilibrium

Forward hourly, spot per-period. We derive the per-product spot best response, work through full and strategic arbitrage to obtain the within-block dispersion equations, and finally compute the within-hour wedge SD in this asymmetric state.

The strategic bidder's Stage 2 problem now has Q spot products. Directly from the spot residual demand $q_{2t} = b_{2t}(p_1 - p_{2t}) + s_t - S_{2t}^{op}$ in (4) and the FOC $q_{2t} = b_{2t}(p_{2t} - c)$, the strategic per-period spot best response is

$$p_{2t}^*(p_1, s_t) \;=\; \frac{p_1 + c}{2} \;+\; \frac{s_t - S_{2t}^{op}}{2\,b_{2t}}. \tag{15}$$

The cross-product variation in p_{2t}^* in this state comes purely from per-subperiod variation in b_{2t}, s_t, S_{2t}^{op} — the forward price p_1 is hourly and shared, and μ_t does not enter the spot demand directly (it acts only through the forward-cleared volume, which is uniform per subperiod when forward is hourly).

D.5.1 Full arbitrage: block parity and within-block dispersion

Block parity requires $p_1 = (1/Q)\sum_t p_{2t}$. Substituting (15),

$$p_1 \;=\; \frac{p_1 + c}{2} + \frac{1}{2Q}\sum_t \frac{s_t - S_{2t}^{op}}{b_{2t}} \quad\Longleftrightarrow\quad p_1 - c \;=\; \frac{1}{Q}\sum_t \frac{s_t - S_{2t}^{op}}{b_{2t}}.$$

Setting $S_{2t}^{op} \equiv 0$ and uniform arbitrage $s_t \equiv \bar{s}$ to isolate the thinness mechanism, and $Q = 2$ for concreteness,

$$p_1 - c \;=\; \frac{\bar{s}}{2}\left(\frac{1}{b_{21}} + \frac{1}{b_{22}}\right) \;=\; \frac{\bar{s}}{2}\cdot\frac{b_{21} + b_{22}}{b_{21}b_{22}} \quad\Longleftrightarrow\quad \bar{s} \;=\; \frac{2(p_1 - c)\,b_{21}b_{22}}{b_{21} + b_{22}}.$$

The within-block dispersion follows from differencing p_{2t}^* across t:

$$p_{21} - p_{22} \;=\; \left(\frac{\bar{s}}{2b_{21}} - \frac{\bar{s}}{2b_{22}}\right) \;=\; \frac{\bar{s}}{2}\cdot\frac{b_{22} - b_{21}}{b_{21}\,b_{22}}.$$

Substituting $\bar{s}$ from above,

$$\boxed{p_{21} - p_{22} \;=\; (p_1 - c)\cdot\frac{b_{22} - b_{21}}{b_{21} + b_{22}}} \qquad \text{(full arbitrage).} \tag{16}$$

D.5.2 Strategic arbitrage: block wedge and attenuated dispersion

The strategic arbitrageur internalises price impact and stops at

$$p_1 - (1/Q)\sum_t p_{2t} \;=\; \frac{p_1 - c}{4}.$$

The block-wedge condition is the average of (1) across the Q matched spot products. Repeating the algebra of §D.5.1 with this modified parity, the arbitrage position $\bar{s}$ falls to exactly half of the full-arbitrage value:

$$\bar{s}^S \;=\; \frac{(p_1 - c)\, b_{21} b_{22}}{b_{21} + b_{22}} \;=\; \frac{\bar{s}^F}{2}.$$

The within-block dispersion correspondingly becomes

$$\boxed{p_{21} - p_{22} \;=\; \frac{p_1 - c}{2} \cdot \frac{b_{22} - b_{21}}{b_{21} + b_{22}}} \qquad \text{(strategic arbitrage)}. \tag{17}$$

The dispersion is attenuated by a factor of two relative to the full-arbitrage case, reflecting the smaller arbitrage volume traded under strategic incentives.

D.5.3 Two-tranche ladder on the spot side

The strategic bidder posts per-product spot ladders. Applying §D.3 with $b \to b_{2t}$ gives ladder spread $\Delta p^*_{2t} = \Delta_\mu/(2b_{2t})$, strictly larger than the spread under hourly spot (which uses $\tilde{\Delta}_\mu = \Delta_\mu/\sqrt{Q}$). This is the spot-side richer-ladder mechanism reported in the main body, eq. (3).

D.5.4 Wedge SD in the asymmetric state

The per-subperiod wedge $w_t = p_1 - p^*_{2t}$ inherits the per-period variation in p^*_{2t}; p_1 is constant across t. From (15), p^*_{2t} varies across t only through per-subperiod thinness b_{2t} (and the arbitrage and aggregator terms when those vary by t). For two matched products ($Q = 2$) the population SD of w_t equals half the within-block price difference, so the full and strategic arbitrage benchmarks give

$$\mathrm{sd}^F_t(w_t) \;=\; \frac{(p_1 - c)\,|b_{22} - b_{21}|}{2(b_{21} + b_{22})}, \qquad \mathrm{sd}^S_t(w_t) \;=\; \frac{(p_1 - c)\,|b_{22} - b_{21}|}{4(b_{21} + b_{22})}, \tag{18}$$

the strategic value exactly half of the full-arbitrage value. Under limited arbitrage with per-product capacity $s_t = K/Q$ binding, the wedge becomes $w_t = (p_1 - c)/2 - K/(2Qb_{2t})$, and

$$\mathrm{sd}^L_t(w_t) \;=\; \frac{K\,|b_{22} - b_{21}|}{8\, b_{21}\, b_{22}} \qquad (Q = 2,\; s_t = K/2), \tag{19}$$

which scales with K rather than with $(p_1 - c)$ but inherits the same $|b_{22} - b_{21}|$ thinness asymmetry.

D.6 (Q, Q) equilibrium

Both markets per-period. We derive the strategic forward best response per product, work through full, strategic, and (the empirically relevant) limited-arbitrage regimes, and close with the Cournot scale-up of cleared MWh and the Stage-1 option-value mechanism.

The strategic per-period forward best response from (4) with $\bar{A}_t = \bar{A} + \mu_t$:

$$p^*_{1t} \;=\; \frac{\bar{A} + \mu_t - S^{op}_{1t} + b_{1t}\, c}{2\, b_{1t}}. \tag{20}$$

D.6.1 Full arbitrage: product parity

Per-product parity $p_{1t} = p_{2t}$ pins s_t at $s_t = b_{2t}(p_{1t} - c) + S^{op}_{2t}$ (same structure as the $(1, 1)$ full-arbitrage condition, but per product). The wedge SD is zero by construction.

D.6.2 Strategic arbitrage: product wedge and SD

Per-product strategic arbitrage gives (1) applied per product: $p_{2t} = (3p_{1t} + c)/4$. The per-product wedge is $p_{1t} - p_{2t} = (p_{1t} - c)/4$. Cross-period SD:

$$\mathrm{sd}_t(p_{1t} - p_{2t}) \;=\; \tfrac{1}{4}\,\mathrm{sd}_t(p^*_{1t}).$$

From (20) with S^{op}_{1t} approximately matched to $\bar{y}_t$:

$$\mathrm{sd}_t(p^*_{1t}) \;=\; \mathrm{sd}_t\!\left(\frac{\mu_t}{2b_1}\right) \;=\; \frac{\sigma_\mu}{2\,b_1}.$$

Therefore

$$\boxed{\mathrm{sd}_t(p_{1t} - p_{2t}) \;=\; \frac{1}{4}\cdot\frac{\sigma_\mu}{2\,b_1} \;=\; \frac{\sigma_\mu}{8\,b_1}.} \tag{21}$$

D.6.3 Two-tranche ladder on the forward side

Applying §D.3 with $b \to b_{1t}$ gives forward-side per-product ladder spread $\Delta p^*_{1t} = \Delta_\mu/(2b_{1t})$, mirroring the spot-side result of §D.5.3.

D.6.4 Limited arbitrage at the symmetric state (empirical regime)

The strategic-arbitrage wedge (21) assumes the arbitrageur can build any per-product position. With Q matched products per clock-hour the arbitrageur's capacity K is spread over Q positions, so the effective per-product capacity is $K_t \approx K/Q$. When this capacity binds — the empirically relevant case in electricity (see §3.6) — the per-product equilibrium is Ito and Reguant Result 3 (limited arbitrage) applied product by product.

Strategic spot best response under binding per-product capacity. Setting $s_t = K_t$ in the strategic Stage 2 FOC (15):

$$p^*_{2t} \;=\; \frac{p^*_{1t} + c}{2} + \frac{K_t - S^{op}_{2t}}{2\,b_{2t}}.$$

The forward per-product cleared price p^*_{1t} depends on μ_t through the forward demand $\bar{A}_t = \bar{A} + \mu_t$; from (20) (corrected to include the arbitrage position), $p^*_{1t} = (\bar{A} + \mu_t - S^{op}_{1t} - s_t + b_{1t}c)/(2b_{1t})$ so $\mathrm{sd}_t(p^*_{1t}) = \sigma_\mu/(2b_1)$ (under uniform $s_t = K_t$ and S^{op}_{1t} matched to structural production). The per-product wedge is

$$p^*_{1t} - p^*_{2t} \;=\; \frac{p^*_{1t} - c}{2} \;-\; \frac{K_t - S^{op}_{2t}}{2\,b_{2t}}.$$

Wedge SD. Taking the cross-product SD:

$$\mathrm{sd}_t(p^*_{1t} - p^*_{2t}) \;=\; \tfrac{1}{2}\,\mathrm{sd}_t(p^*_{1t}) \;=\; \frac{\sigma_\mu}{4\,b_1}.$$

Compare to the strategic-arbitrage closed-form (21): under unlimited strategic arbitrage the wedge is $(p_{1t} - c)/4$ giving wedge SD $\sigma_\mu/(8\,b_1)$; under limited arbitrage the wedge is $(p_{1t} - c)/2$ giving wedge SD $\sigma_\mu/(4\,b_1)$, exactly twice the strategic-arbitrage value. The wedge SD does *not* collapse fully even at the symmetric state. If per-period thinness b_{2t} also varies across t (as in the asymmetric state) the wedge inherits an additional thinness-asymmetry component on top of $\sigma_\mu/(4b_1)$; the empirically observed sustaining of wedge SD at granular-state magnitudes is consistent with this combined effect.

Per-product capacity binding. For the limited regime to bind, K_t must be smaller than the strategic-arbitrage equilibrium $s_t^{\text{strat}} = b_{2t}(p_{1t} - c)/2$. With $K_t \approx K/Q$ and s_t^{strat} proportional to a per-product mark-up, binding requires $K \leq Q \cdot \bar{s}^{\text{strat}}$ where $\bar{s}^{\text{strat}}$ is the average per-product strategic position. Empirically, hourly arbitrage capacity is set by participation rules and balance-sheet limits that do not scale with Q when granularity reforms multiply products; the constraint binds at the granular state by construction.

D.6.5 *Cournot cleared-MWh under per-period thinness*

The strategic best response (20) gives per-period cleared quantity $q_{1t}^* = (\bar{A}_t - b_{1t}c - S_{1t}^{op} - s_t)/2$. Summing across t and taking expectation over μ_t:

$$\mathbb{E}\Big[\sum_t q_{1t}^*\Big] \;=\; \frac{Q\bar{A} - c\sum_t b_{1t} - \sum_t S_{1t}^{op} - \sum_t s_t}{2}.$$

The hourly counterpart is $Q \cdot q_1^*|_{(1,1)} = Q \cdot (\bar{A} - b_1 c - S_1^{op} - s)/2$. Comparing:

$$\mathbb{E}\Big[\sum_{t=1}^{Q} q_{1t}^*\Big]\Big|_{(Q,Q)} \;-\; Q \cdot q_1^*\big|_{(1,1)} \;=\; \frac{c}{2}\left(Q\,b_1 - \sum_{t=1}^{Q} b_{1t}\right). \tag{22}$$

Positive whenever $\sum_{t=1}^{Q} b_{1t} < Q\,b_1$ — i.e. when per-period thinness is on average smaller than the hourly aggregate elasticity, as expected (each 15-min product is thinner than the hourly aggregate it replaces). This is the Cournot scale-up of main-body §3.5(iv), obtained without any imported mechanism.

D.6.6 *Stage-1 option value (cross-market substitution)*

Per-period spot Cournot delivers a Stage-2 profit $\pi_2(p_1, \mu_t)$ that is exactly quadratic in μ_t (the per-period FOC is linear in μ_t, the equilibrium quantity is linear, and the profit is the product). Stage-1 expected profit then equals

$$\mathbb{E}_{\mu_t}[\Pi(p_1)] \;=\; \pi_1(p_1) \;+\; \pi_2(p_1, 0) \;+\; \tfrac{1}{2}\,\mathbb{E}[\mu_t^2]\,\frac{\partial^2 \pi_2}{\partial \mu_t^2}\bigg|_{\mu_t=0}$$

exactly (no higher-order remainder, because $\partial^k \pi_2/\partial \mu_t^k = 0$ for $k \geq 3$). Under the uniform shock $\mu_t \sim U[-\Delta_\mu, \Delta_\mu]$ used throughout the welfare derivations, $\mathbb{E}[\mu_t^2] = \Delta_\mu^2/3$ and $\partial^2\pi_2/\partial\mu_t^2|_0 = 1/(2b_{2t})$, so the variance correction equals $\Delta_\mu^2/(12b_{2t})$.

The variance correction is strictly positive: the Stage-1 bidder optimises against a more valuable Stage-2 option, widening the forward ladder to capture both sides of the variation. Symmetrically when the forward is per-period: the within-hour discrimination is absorbed at Stage 1 directly, so the Stage-2 option value drops and the spot ladder tightens. This is the model-internal account of the cross-market substitution pattern of §6.1.

D.7 Balancing discrepancy

The model has two disjoint balancing channels and we keep them separate throughout. The first is the *individual* imbalance cost: each participant pays for its own deviation from schedule. The aggregator internalises its own production shock η_t through the penalty $\gamma \cdot \mathbb{E}[(y_t - S_t^{op})^2]$ already accounted for in $\pi^{Agg,\text{net}}$ (§D.2). The strategic bidder produces with certainty, so its individual imbalance is identically zero. The second is the *system-side* discrepancy on the demand leg: the within-hour demand shock μ_t that no hourly market absorbs, and that no individual contract internalises. The discrepancy Δ_t below collects both channels for the system view that enters C^I_{sys} in the welfare derivations (§D.9); the aggregator's individual imbalance is the $(y_t - S^{op}_t)$ component, and the demand-side μ_t is the part with no individual owner.

With $\bar{A}_t = \bar{A} + \mu_t$ and aggregator realisation $y_t = \bar{y}_t + \eta_t$,

$$\Delta_t \;=\; (\bar{A}_t - q^{\text{cleared}}_t) \;+\; (y_t - S^{op}_t) \;+\; \varepsilon_t, \tag{23}$$

where $S^{op}_t = S^{op}_{1t} + S^{op}_{2t}$ is the aggregator's total scheduled MWh at subperiod t and ε_t is residual noise unrelated to μ_t and η_t.

State $(1,1)$. Forward and spot are both hourly: under the structural assumption $\bar{y}_t \equiv \bar{Y}_h/Q$ (no within-hour structural production variation), $\Delta_t = \mu_t + \eta_t + \varepsilon_t$.

State $(1,Q)$. The spot becomes per-period: the aggregator re-schedules at Stage 2 using its updated forecast, absorbing the η_t shock. The forward remains hourly so μ_t is still unabsorbed: $\Delta_t = \mu_t + \varepsilon_t$.

State (Q,Q). The forward also becomes per-period: per-product forward clearing q_{1t} responds to $\bar{A}_t = \bar{A}+\mu_t$ through (20), absorbing μ_t. The aggregator's η_t remains absorbed at spot per-period. The balancing discrepancy reduces to $\Delta_t = \varepsilon_t$.

These reductions are mechanical in the contracting grid and hold under any arbitrage regime.

D.8 Note on the alternative sequence

The mirror path $(1,1) \to (Q,1) \to (Q,Q)$ swaps the order of the two auction reforms. After relabelling, the same per-state algebra of §D.4–§D.6 applies. The end-state (Q,Q) is identical. The intermediate state $(Q,1)$ produces a within-hour *DA* dispersion driven by per-subperiod b_{1t} asymmetry (mirror of (16)–(17) but with roles of b_{1t} and b_{2t} swapped); the aggregator's quantity-shift mechanism activates already at the forward reform; the balancing-discrepancy gain $\Delta_t : \mu_t + \varepsilon_t \to \varepsilon_t$ realises at the forward reform under either path. The peak wedge SD ($\sigma_\mu/(2\,b_2)$ in the original path, $\sigma_\mu/(2\,b_1)$ in the mirror) and the final wedge SD ($\sigma_\mu/(8\,b_1)$ in both) are sequence-invariant up to the relabelling of the dominant elasticity.

D.9 Welfare derivations

We collect the closed-form welfare components by state and arbitrage regime under uniform $\mu_t \sim [-\Delta_\mu, \Delta_\mu]$ and $\eta_t \sim [-\Delta_\eta, \Delta_\eta]$ (residual noise ε_t negligible). The aggregator's penalty enters $\pi^{Agg,\text{net}}$ through η_t and the system imbalance cost C^I_{sys} enters through μ_t; the two channels are disjoint.

D.9.1 Components

Per clock-hour:

$$W \;=\; CS \;+\; \pi^{SB} \;+\; \pi^{Agg,\text{net}} \;+\; \pi^{Arb} \;-\; C^I_{\text{sys}}. \tag{24}$$

The aggregator's net profit $\pi^{Agg,\text{net}}$ is its revenue $\sum_t (p^{(t)}_1 S^{op}_{1t} + p_{2t} S^{op}_{2t})$ minus its internalised quadratic penalty $\gamma\, E[\mathcal{I}^{Agg}] = \gamma \sum_t E[(\eta^{\text{unhedged}}_t)^2]$ (§D.2). The system imbalance cost is $C^I_{\text{sys}} = \gamma \sum_t E[\Delta_t^2]$ with Δ_t from §D.7. $\pi^{Arb} = s(p_1 - p_2)$ is the arbitrageur's revenue ($\pi^{Arb} = 0$ under no arbitrage and full arbitrage). For linear residual demand $q = \bar{A} - bp$, consumer surplus at p^* is

$$CS \;=\; \frac{(\bar{A} - bp^*)^2}{2b}. \tag{25}$$

D.9.2 Imbalance cost decomposition (closed form, uniform shocks)

Under $\mu_t \sim U[-\Delta_\mu, \Delta_\mu]$ and $\eta_t \sim U[-\Delta_\eta, \Delta_\eta]$ independent (and $\varepsilon_t \approx 0$): $E[\mu_t^2] = \Delta_\mu^2/3$, $E[\eta_t^2] = \Delta_\eta^2/3$. The aggregator's unhedged shock is η_t when spot is hourly and 0 once spot is per-period; the system discrepancy is μ_t when forward is hourly and 0 once forward is per-period. The quadratic-penalty costs decompose as

	$(1,1)$	$(1,Q)$	(Q,Q)
$\gamma\, E[\mathcal{I}^{Agg}]$ (in $\pi^{Agg,\text{net}}$)	$\gamma Q\Delta_\eta^2/3$	0	0
C^I_{sys} (separate)	$\gamma Q\Delta_\mu^2/3$	$\gamma Q\Delta_\mu^2/3$	0
Total imbalance cost	$\gamma Q(\Delta_\mu^2+\Delta_\eta^2)/3$	$\gamma Q\Delta_\mu^2/3$	0

(26)

The two channels are disjoint (η_t only enters $\pi^{Agg,\text{net}}$, μ_t only enters C^I_{sys}).

D.9.3 Welfare at state $(1,1)$ by arbitrage regime

No arbitrage (R1). From §D.4.2: $p_1^{NA} = (\bar{A} + b_1 c - S_1^{op})/(2b_1)$, $q_1^{NA} = (\bar{A} - b_1 c - S_1^{op})/2$, $\pi_1^{SB,NA} = (\bar{A} - b_1 c - S_1^{op})^2/(4b_1)$. Consumer surplus at the forward stage is, by (25), $CS_1^{NA} = (\bar{A} - b_1 p_1^{NA})^2/(2b_1) = (q_1^{NA} + S_1^{op})^2/(2b_1)$.

Full arbitrage (R2). At $p_1 = p_2 = p_1^F = (\bar{A} - \bar{Y}_h + b_1 c)/(2b_1)$ from (13), strategic-bidder profit is $\pi^{SB,F} = (\bar{A} - b_1 c - \bar{Y}_h)^2/(4b_1)$. The cleared MWh stays below the competitive benchmark (Ito and Reguant Result 2): with monopoly market power, even competitive arbitrage cannot restore the first-best.

Strategic arbitrage (R4). The arbitrageur extracts $\pi^{Arb,S} = b_2(p_1 - c)^2/4$, a pure transfer that subtracts from the strategic bidder's rent.

Limited arbitrage (R3, empirical). With $s = K < b_2(p_1 - c)/2$ binding, the arbitrageur extracts $\pi^{Arb,L} = K\,[(p_1 - c)/2 - (K - S_2^{op})/(2b_2)]$.

D.9.4 Welfare at $(1,Q)$ and (Q,Q) — delta from $(1,1)$

Spot reform, $(1,1) \to (1,Q)$: aggregator hedging. $E[\mathcal{I}^{Agg}]$ drops from $Q\,\Delta_\eta^2/3$ to 0, so $\Delta\pi^{Agg,\text{net}} = +\gamma Q\,\Delta_\eta^2/3$.

Spot-side ladder rent. The strategic bidder's spot-side two-tranche ladder (3) replaces a single-price Cournot with a Δ_μ-discriminated ladder. By §D.3 closed form:

$$\Delta\pi^{SB,2T}_{\text{spot}} \;=\; \frac{\Delta_\mu\left[4(\bar{A} - b_{2t}c) + \Delta_\mu\right]}{16\,b_{2t}}. \tag{27}$$

On the inframarginal volume this is a CS-to-π^{SB} transfer; on the marginal extension this is new surplus.

Forward reform, $(1,Q) \to (Q,Q)$: system imbalance. Per-product forward absorbs μ_t, so C^I_{sys} drops from $\gamma Q\,\Delta_\mu^2/3$ to 0:

$$\Delta(-C^I_{\text{sys}}) \;=\; +\gamma Q\,\Delta_\mu^2/3.$$

Forward-side ladder rent.

$$\Delta\pi^{SB,2T}_{\text{forward}} \;=\; \frac{\Delta_\mu\left[4(\bar{A} - b_1 c) + \Delta_\mu\right]}{16\,b_1}. \tag{28}$$

Cournot scale-up at (Q, Q)**.** Total cleared MWh per clock-hour rises by $(c/2)(Qb_1 - \sum_t b_{1t})$ via (22). At the mark-up $p_1 - c = (\bar{A} - b_1 c - S_1^{op})/(2b_1)$,

$$\Delta W^{\text{scale}} = \frac{c\,(\bar{A} - b_1 c - S_1^{op})}{4\,b_1}\Big(Qb_1 - \sum_t b_{1t}\Big). \tag{29}$$

Limited-arbitrage gain at $(1, Q)$ **and** (Q, Q)**.** Per-product capacity K/Q binds, leaving a per-product wedge $(p_{1t} - c)/2$ instead of the unlimited-strategic $(p_{1t} - c)/4$ (§D.6.4). The strategic bidder's spot markup falls from $3(p_1 - c)/4$ to $(p_1 - c)/2$, so the per-product deadweight loss falls by $5b_2(p_1 - c)^2/32$ (Ito and Reguant (2016) Result 2). This is the welfare counterpart of the empirical asymmetric-persistence wedge-SD finding — not a welfare cost.

D.9.5 Net welfare deltas (closed form)

Combining the channels for each reform step (writing $a_m \equiv \bar{A} - b_m c$ for $m \in \{1, 2\}$):

$$\Delta W^{\text{spot}} = \frac{\gamma Q \Delta_\eta^2}{3} + \frac{5\,b_2(p_1 - c)^2}{32} + \frac{\Delta_\mu\,[4a_2 + \Delta_\mu]}{16\,b_2}, \tag{30}$$

$$\Delta W^{\text{forward}} = \frac{\gamma Q \Delta_\mu^2}{3} + \frac{c\,a_1}{4\,b_1}\Big(Qb_1 - \sum_t b_{1t}\Big) + \frac{\Delta_\mu\,[4a_1 + \Delta_\mu]}{16\,b_1}, \tag{31}$$

where the second term of (30) is the per-product limited-arbitrage gain at the regime change, and the third terms in both lines are the bid-ladder rent extensions (mostly distributional).

Sign and dominance. Both deltas are unconditionally positive — the model predicts welfare improvement at each reform step, with imbalance saving the dominant channel given that balancing prices in Spanish electricity markets exceed day-ahead clearing prices by a factor of 2–10.